

Fast and Robust Phase Equilibrium Computations for $\text{CO}_2 + \text{H}_2\text{O} + \text{NaCl}$ Mixtures Using the Electrolyte Cubic-Plus-Association Equation of State

Supplementary Information

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This document provides supplementary material for the main paper. Section S1 gives a complete nomenclature listing. Section S2 tabulates the CPA/eCPA equation-of-state parameters and the coefficients of the temperature-dependent Péneloux volume shift for H_2O introduced in this work. Section S3 derives the closed-form analytical Jacobian matrices used by the inner Newton–Raphson solvers for the CPA and eCPA phase-fugacity calculations. Section S4 presents additional validation figures for $\text{CO}_2 + \text{H}_2\text{O}$ and $\text{CO}_2 + \text{NaCl}$ phase equilibria, aqueous density, and the simplified flash. Section S5 shows supplementary convergence and efficiency figures. Section S6 provides the full reservoir-simulation setup and performance tables.

S1. Nomenclature

This section provides a complete listing of all mathematical symbols used throughout the paper. Latin symbols are given in Table S1; Greek symbols, subscripts, and superscripts follow in the unnumbered tables below. Within each group, symbols are listed in alphabetical order.

Latin symbols

Table S1: Nomenclature: all mathematical symbols used in this work. Latin symbols are listed below; Greek symbols, subscripts, and superscripts follow in the unnumbered tables at the end of this section.

Symbol	Description	Units
<i>Upper case</i>		
A	Dimensionless SRK attractive parameter, $A = a(T)P/(RT)^2$	—
A^r	Residual Helmholtz energy per mole	J mol^{-1}
A^{assoc}	Wertheim association term of A^r	J mol^{-1}
A^{Born}	Born solvation contribution to A^r	J mol^{-1}

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Table S1 (*continued*)

Symbol	Description	Units
A^{DH}	Debye–Hückel electrostatic contribution to A^r	J mol^{-1}
A^{SRK}	Soave–Redlich–Kwong contribution to A^r	J mol^{-1}
AARE	Average absolute relative error	%
B	Dimensionless SRK co-volume parameter, $B = bP/(RT)$	—
D_c	CO_2 association denominator	—
D_w	H_2O association denominator	—
F_1, F_2, F_3	CPA inner-solve residuals in $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$	—
F_z	Molar fractional flow of CO_2	—
G^{res}	Residual Gibbs energy	J mol^{-1}
G_0, G_1, G_2	eCPA inner-solve residuals in $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$	—
H_{ij}	Jacobian entries for the eCPA $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ system	various
J_{ij}	Jacobian entries for the CPA $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ system	various
K	Isotropic permeability	mD
K_i	Vapour–liquid K-value of species i , $K_i = y_i/x_i$	—
K_i^W	Wilson K-value correlation	—
L	Step-size clipping limit in accelerated SSI	—
L_x, L_y, L_z	Reservoir domain dimensions	m
M_w	Molar mass of water ($0.018015 \text{ kg mol}^{-1}$)	kg mol^{-1}
N_A	Avogadro number	mol^{-1}
N_x, N_y	Number of grid cells in x - and y -direction	—
P	Pressure	bar
P_c	Critical pressure	bar
P_{BHP}	Bottom-hole pressure	bar
P_{sat}	Saturation pressure of pure H_2O	bar
R	Universal gas constant	$\text{J mol}^{-1} \text{ K}^{-1}$
R_B	Born radius of ion	$\text{\AA}$
S_g	CO_2 -rich phase saturation	—
$S_{\text{CO}_2, \text{H}_2\text{O}}$	CO_2 – H_2O cross-association strength ratio	—
S_{gr}	Residual gas saturation (Corey)	—
S_{wc}	Connate water saturation (Corey)	—
T	Temperature	$^{\circ}\text{C}$ or K
T_c	Critical temperature	K
TPD(W)	Tangent-plane distance function	—
V_g	Radial distribution function derivative with respect to η	—
W	Second derivative of V_g ; also unnormalized trial composition vector	—
Z	Compressibility factor, $Z = Pv/(RT)$	—
Z^c	Compressibility factor of the CO_2 -rich phase	—
Z^{aq}	Compressibility factor of the aqueous phase	—
<i>Lower case</i>		

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Table S1 (*continued*)

Symbol	Description	Units
$a(T)$	Mixture SRK attractive parameter	$\text{J cm}^3 \text{ mol}^{-2}$
a_{0i}	Pure-component reference attractive parameter	$\text{J cm}^3 \text{ mol}^{-2}$
b	Mixture SRK co-volume parameter	$\text{cm}^3 \text{ mol}^{-1}$
b_i	Pure-component co-volume	$\text{cm}^3 \text{ mol}^{-1}$
c_i	P��neloux volume-shift parameter of species i	$\text{cm}^3 \text{ mol}^{-1}$
c_{1i}	Soave temperature-dependence coefficient of species i	—
d_i	Reference chemical-potential offset in TPD, $\ln z_i + \ln \hat{\phi}_i(\mathbf{z})$	—
$g(\eta)$	Carnahan–Starling radial distribution function	—
g_w	Kirkwood g -factor for water	—
$\mathbf{g}^{(k)}$	Fugacity-ratio residual vector at SSI iteration k	—
k_{ij}	Binary interaction parameter between species i and j	—
k_B	Boltzmann constant	J K^{-1}
k_{rg}, k_{rw}	Corey relative permeabilities (CO ₂ -rich, aqueous)	—
$m^{(k)}$	Dominant-eigenvalue acceleration factor at iteration k	—
m_{CO_2}	CO ₂ molality in the aqueous phase	mol kg^{-1}
m_s	NaCl feed molality (solution-table grid coordinate); no superscript = feed value	mol kg^{-1}
m_s^{aq}	Self-consistent equilibrium NaCl molality in the aqueous phase	mol kg^{-1}
n_g, n_w	Corey exponents (CO ₂ -rich, aqueous)	—
q	Well volumetric flow rate	$\text{m}^3 \text{ s}^{-1}$
r_w	Wellbore radius	m
t	Time	yr
v	Molar volume	$\text{cm}^3 \text{ mol}^{-1}$
w_i	Normalised trial mole fractions in stability test	—
x_i	Mole fraction of species i in the aqueous phase	—
$x_{\text{CO}_2}^{\text{sf}}$	CO ₂ mole fraction on a salt-free basis	—
y_i	Mole fraction of species i in the CO ₂ -rich phase	—
$\mathbf{z}$	Overall feed composition vector, $\mathbf{z} = (z_{\text{H}_2\text{O}}, z_{\text{CO}_2})$	—
$z_{\text{H}_2\text{O}}$	Overall H ₂ O feed mole fraction, $z_{\text{H}_2\text{O}} = 1 - z_{\text{CO}_2}$	—
z_{CO_2}	Overall CO ₂ feed mole fraction	—

Greek symbols

Symbol	Description	Units
α_{is}	NRTL non-randomness parameter (ion–solvent)	—
β	CO ₂ -rich phase molar fraction	—
$\delta^{A_i B_j}$	Association bonding volume between sites A_i and B_j	m ³ mol ⁻¹
ΔU_{is}	Ion–solvent interaction energy (eCPA NRTL term)	J mol ⁻¹
ε/k_B	Association well depth (temperature units)	K
ε_0	Vacuum permittivity	F m ⁻¹
ε_∞	High-frequency (optical) relative permittivity	—
ε_r	Static relative permittivity (dielectric constant) of water	—
η	Packing fraction, $\eta = b/(4v)$	—
$\hat{\phi}_i$	Fugacity coefficient of species i	—
κ	Association volume parameter in the Wertheim term	—
μ_{0w}	Permanent dipole moment of water	C m
μ_g, μ_w	Dynamic viscosity (CO ₂ -rich, aqueous)	Pa s
ρ	Mass density	kg m ⁻³
ρ_N	Molar density (mixture), $\rho_N = P/(ZRT)$	mol m ⁻³
σ	Collision diameter (ion hard-sphere radius)	Å
$\tau, \tau_{\text{switch}}$	SSI convergence tolerance; Newton-polish switch threshold	—
ϕ	Porosity	—
χ^{A_i}	Fraction of molecules of species i not bonded at site A	—
$\chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2}$	Non-bonded fractions of H ₂ O and CO ₂	—
ω_i	Acentric factor of species i	—

Subscripts and superscripts

Symbol	Meaning
c	CO ₂ -rich phase
aq	Aqueous phase
i, j	Species index
H ₂ O, CO ₂	Component (H ₂ O or CO ₂)
assoc	Association contribution
Born	Born solvation contribution
DH	Debye–Hückel electrostatic contribution
feed	Feed (overall) quantity
phys	Physical (SRK) contribution
res	Residual thermodynamic quantity
SRK	Soave–Redlich–Kwong reference fluid
c, i	Critical property of species i

S2. EoS Parameters

This section supplements the equation-of-state section of the main text. For completeness, Tables S2 and S3 reproduce the key EoS parameters used in this work, all from Coelho et al. [1]. The coefficients

of the temperature-dependent P  neloux volume shift for H₂O (Eq. S1), introduced in this work, are given in Table S4.

Table S2: Pure-component CPA/eCPA parameters [1]. For ions, $a_0 = 0$. H₂O uses the 4C association scheme.

Species	a_0 (J m ³ mol ⁻²)	c_1	b (cm ³ mol ⁻¹)	ε/k_B (K)	κ	σ (��)	R_B (��)
H ₂ O	0.1228	0.6736	14.52	2003	κ^*	—	—
CO ₂	0.3508	0.7602	27.20	—	—	—	—
Na ⁺	0	—	16.49	—	—	2.356	1.665
Cl ⁻	0	—	40.83	—	—	3.187	1.828

* $\kappa = \kappa_{\text{assoc}} \cdot b_{\text{H}_2\text{O}}$, $\kappa_{\text{assoc}} = 69.2 \times 10^{-3}$.

Table S3: Temperature-dependent binary interaction parameters [1]. $X = A_X(T/T_{c,\text{CO}_2})^2 + B_X(T/T_{c,\text{CO}_2}) + C_X$, $T_{c,\text{CO}_2} = 31.3^\circ\text{C}$.

Parameter	A	B	C
$k_{\text{CO}_2,\text{H}_2\text{O}}$ (CO ₂ –H ₂ O binary interaction parameter)	-0.492	2.101	-1.571
$S_{\text{CO}_2,\text{H}_2\text{O}}$ (cross-assoc. strength)	+0.192	-0.173	-0.009

Ion–neutral pair	ΔU^{ref} (J mol ⁻¹)	α (K)	T^α (K)	Source
H ₂ O–NaCl (ΔU_{ws})	-1858	1573	340	Maribo-Mogensen et al. (2015)
CO ₂ –NaCl (ΔU_{cs})	+6056	692	244	Coelho et al. [1]

Temperature-dependent P  neloux shift for H₂O

A temperature-dependent volume shift for H₂O is introduced in this work. We fit a degree-4 polynomial in the reduced temperature $T_R = T/T_{c,\text{H}_2\text{O}}$ ($T_{c,\text{H}_2\text{O}} = 647.29\text{ K}$) to 455 IAPWS-95 liquid/compressed-water conditions ($T = 0\text{--}350^\circ\text{C}$, $P = 1\text{--}2000\text{ bar}$):

$$c_{\text{H}_2\text{O}}(T) = a_4 T_R^4 + a_3 T_R^3 + a_2 T_R^2 + a_1 T_R + a_0 \quad [\text{cm}^3 \text{ mol}^{-1}]. \quad (\text{S1})$$

The fitted coefficients are given in Table S4.

Table S4: Coefficients of the degree-4 P  neloux volume shift polynomial for H₂O (Eq. S1), fitted in this work to 455 IAPWS-95 liquid/compressed-water conditions ($T = 0\text{--}350^\circ\text{C}$, $P = 1\text{--}2000\text{ bar}$).

Coefficient	Value
a_4	+32.862
a_3	-96.052
a_2	+110.954
a_1	-58.111
a_0	+11.233

S3. Analytical Jacobians for the Inner Newton Solvers

This section supplements the SSI and K-value flash sections of the main text, which describe the inner Newton–Raphson solvers used at each SSI iteration to evaluate the CPA and eCPA phase fugacities. Here we derive the full analytical Jacobian matrices that make those Newton steps computationally efficient. Within each SSI iteration two distinct Newton systems are solved, one per phase. The CO₂-rich phase and both phases of the salt-free CPA binary use a 3×3 system in $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$ (Section S3.1). The eCPA aqueous phase requires a different 3×3 system in $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$, because the relative permittivity ε_r satisfies an additional Kirkwood self-consistency equation (Section S3.2).

S3.1. CPA binary: Newton system for $(Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})$

The system is written for the aqueous phase (mole fractions x_i); the CO₂-rich phase uses the identical equations with y_i in place of x_i . The three fixed-point residuals are

$$F_1 = Z - \frac{Z}{Z - B} + \frac{A}{Z + B} - G[x_{\text{H}_2\text{O}}(\chi_{\text{H}_2\text{O}} - 1) + x_{\text{CO}_2}(\chi_{\text{CO}_2} - 1)] = 0, \quad (\text{S2})$$

$$F_2 = \chi_{\text{H}_2\text{O}} - \frac{Z}{D_w} = 0, \quad (\text{S3})$$

$$F_3 = \chi_{\text{CO}_2} - \frac{Z}{D_c} = 0, \quad (\text{S4})$$

with the denominators

$$D_w = Z + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}\Delta_w + 2x_{\text{CO}_2}\chi_{\text{CO}_2}\Delta_{\text{CO}_2, \text{H}_2\text{O}}, \quad D_c = Z + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}\Delta_{\text{CO}_2, \text{H}_2\text{O}}, \quad (\text{S5})$$

where $\Delta_w = g(\eta)\kappa_w[\exp(\varepsilon_w/k_B T) - 1]P/(RT)$, $\Delta_{\text{CO}_2, \text{H}_2\text{O}} = S_{\text{CO}_2, \text{H}_2\text{O}}\Delta_w$, $\eta = B/(2Z)$, and $G = 1 + V_g$ with

$$V_g \equiv \frac{\eta g'(\eta)}{g(\eta)} = \frac{\eta(5 - 2\eta)}{(1 - \eta)(2 - \eta)}, \quad W \equiv \frac{dV_g}{d\eta} = \frac{10 - 8\eta + \eta^2}{(1 - \eta)^2(2 - \eta)^2}, \quad (\text{S6})$$

for the Carnahan–Starling radial distribution function $g(\eta) = (2 - \eta)/[2(1 - \eta)^3]$. Because $\eta \propto Z^{-1}$, the key identity is

$$Z \frac{\partial \Delta_w}{\partial Z} = -V_g \Delta_w, \quad (\text{S7})$$

from which all Z -derivatives of D_w and D_c follow. Setting $D = x_{\text{H}_2\text{O}}(\chi_{\text{H}_2\text{O}} - 1) + x_{\text{CO}_2}(\chi_{\text{CO}_2} - 1)$, the 3×3 Jacobian $J_{ij} = \partial F_i / \partial v_j$ with $\mathbf{v} = (Z, \chi_{\text{H}_2\text{O}}, \chi_{\text{CO}_2})^\top$ is:

Row 1.

$$J_{11} = 1 + \frac{B}{(Z - B)^2} - \frac{A}{(Z + B)^2} + \frac{DW\eta}{Z}, \quad (\text{S8})$$

$$J_{12} = -G x_{\text{H}_2\text{O}}, \quad (\text{S9})$$

$$J_{13} = -G x_{\text{CO}_2}. \quad (\text{S10})$$

Row 2. Using $Z\partial D_w/\partial Z = Z - 2(x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}} + x_{\text{CO}_2}\chi_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}})V_g\Delta_w$ and $\chi_{\text{H}_2\text{O}} = Z/D_w$:

$$J_{21} = -\frac{2\Delta_w(1+V_g)(x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}} + x_{\text{CO}_2}\chi_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}})}{D_w^2}, \quad (\text{S11})$$

$$J_{22} = 1 + \frac{2x_{\text{H}_2\text{O}}\Delta_w\chi_{\text{H}_2\text{O}}}{D_w}, \quad (\text{S12})$$

$$J_{23} = \frac{2x_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w\chi_{\text{H}_2\text{O}}}{D_w}. \quad (\text{S13})$$

Row 3. Using $Z\partial D_c/\partial Z = Z - 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}S_{\text{CO}_2,\text{H}_2\text{O}}V_g\Delta_w$ and $\chi_{\text{CO}_2} = Z/D_c$:

$$J_{31} = -\frac{2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w(1+V_g)}{D_c^2}, \quad (\text{S14})$$

$$J_{32} = \frac{2x_{\text{H}_2\text{O}}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w\chi_{\text{CO}_2}}{D_c}, \quad (\text{S15})$$

$$J_{33} = 1. \quad (\text{S16})$$

Note $J_{33} = 1$ because D_c does not depend on χ_{CO_2} ; Newton therefore has a direct update for χ_{CO_2} at each step. For the CO_2 -rich phase the identical Jacobian applies with y_i in place of x_i .

S3.2. eCPA aqueous phase: Newton system for $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$

In the aqueous phase the relative permittivity ε_r is an additional unknown satisfying the Kirkwood self-consistency equation (S18). The CO_2 cross-association fraction χ_{CO_2} is eliminated algebraically via $\chi_{\text{CO}_2} = Z^{\text{aq}}/D_c$ with $D_c = Z^{\text{aq}} + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w$, reducing the system to three equations in $\mathbf{v} = (Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})^{\top}$:

$$G_0 = Z^{\text{aq}} - Z^{\text{aq,phys}} - Z^{\text{aq,assoc}} - Z^{\text{aq,DH}} - Z^{\text{aq,Perm}} = 0, \quad (\text{S17})$$

$$G_1 = T_1(\varepsilon_r, \varepsilon_{\infty}) - T_2(Z^{\text{aq}}, \chi_{\text{H}_2\text{O}}) = 0, \quad (\text{S18})$$

$$G_2 = \chi_{\text{H}_2\text{O}} - \frac{Z^{\text{aq}}}{D_w} = 0, \quad (\text{S19})$$

where T_1 and T_2 are the left- and right-hand sides of the Onsager–Kirkwood relative permittivity fixed-point equation [1], $D_w = Z^{\text{aq}} + 2x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}}\Delta_w + 2x_{\text{CO}_2}\chi_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}}\Delta_w$, and all Z^{aq} -derivatives of Δ_w satisfy the same identity (S7) (with Z^{aq} in place of Z).

Row 2 (exact). G_2 has the same structure as F_2 (S3) with ε_r absent from D_w , so

$$H_{20} = -\frac{2\Delta_w(1+V_g)(x_{\text{H}_2\text{O}}\chi_{\text{H}_2\text{O}} + x_{\text{CO}_2}\chi_{\text{CO}_2}S_{\text{CO}_2,\text{H}_2\text{O}})}{D_w^2}, \quad (\text{S20})$$

$$H_{21} = 0, \quad (\text{S21})$$

$$H_{22} = 1 + \frac{2x_{\text{H}_2\text{O}}\Delta_w\chi_{\text{H}_2\text{O}}}{D_w}. \quad (\text{S22})$$

Row 1 (*exact*). $T_1 = F(\varepsilon_r, \varepsilon_\infty)$ (left-hand side of the Onsager–Kirkwood equation) and

$$T_2 = (N_A c / 9 \varepsilon_0 k_B T) x_{\text{H}_2\text{O}} g_w \mu_{0w}^2$$

depend on $(Z^{\text{aq}}, \varepsilon_r, \chi_{\text{H}_2\text{O}})$ through $\varepsilon_\infty(Z^{\text{aq}})$, ε_r , and the Kirkwood g-factor $g_w(\chi_{\text{H}_2\text{O}})$, respectively:

$$H_{10} = \left(\frac{\partial F}{\partial \varepsilon_\infty} \right) \frac{V(\partial \varepsilon_\infty / \partial V)}{Z^{\text{aq}}} - \frac{N_A \mu_{0w}^2}{9 \varepsilon_0 k_B T} x_{\text{H}_2\text{O}} \left(-\frac{\rho_N}{Z^{\text{aq}}} g_w + \rho_N \frac{\partial g_w}{\partial Z^{\text{aq}}} \right), \quad (\text{S23})$$

$$H_{11} = \frac{\partial F}{\partial \varepsilon_r} = \frac{2\varepsilon_r^2 + \varepsilon_\infty^2}{\varepsilon_r^2(\varepsilon_\infty + 2)^2}, \quad (\text{S24})$$

$$H_{12} = -\frac{N_A \rho_N \mu_{0w}^2}{9 \varepsilon_0 k_B T} x_{\text{H}_2\text{O}} \frac{\partial g_w}{\partial \chi_{\text{H}_2\text{O}}}, \quad (\text{S25})$$

where $\rho_N = P/(Z^{\text{aq}}RT)$, $V(\partial \varepsilon_\infty / \partial V)$ follows from the Clausius–Mossotti relation for ε_∞ [1], and $\partial g_w / \partial Z^{\text{aq}}$ and $\partial g_w / \partial \chi_{\text{H}_2\text{O}}$ follow from differentiating the Kirkwood g-factor [1] through $\mathcal{P}_{ww}$ and $\mathcal{P}_{wc}$ using Eq. (S7) for $\partial \delta_w / \partial Z^{\text{aq}}$ and $\partial \chi_{\text{CO}_2} / \partial \chi_{\text{H}_2\text{O}} = -Z^{\text{aq}} \cdot 2x_{\text{H}_2\text{O}} S_{\text{CO}_2, \text{H}_2\text{O}} \Delta_w / D_c^2$, respectively.

Row 0 (*partially approximate*). The Debye screening length satisfies

$$\frac{\partial \kappa}{\partial Z^{\text{aq}}} = -\frac{\kappa}{2Z^{\text{aq}}}, \quad \frac{\partial \kappa}{\partial \varepsilon_r} = -\frac{\kappa}{2\varepsilon_r}. \quad (\text{S26})$$

The ε_r -column entry combines DH and permittivity contributions,

$$H_{01} = -\frac{\partial Z^{\text{aq, DH}}}{\partial \varepsilon_r} - \frac{\partial Z^{\text{aq, Perm}}}{\partial \varepsilon_r}, \quad (\text{S27})$$

where $\partial Z^{\text{aq, DH}} / \partial \varepsilon_r$ is obtained from the DH compressibility expression [1] through $\kappa(\varepsilon_r)$ using Eq. (S26), and $\partial Z^{\text{aq, Perm}} / \partial \varepsilon_r$ requires differentiating $Z^{\text{aq, Perm}} = -a_{\varepsilon_r} V(\partial \varepsilon_r / \partial V)$ with respect to ε_r through κ and through $V(\partial \varepsilon_r / \partial V)$, using the second-order F -derivatives

$$\frac{\partial^2 F}{\partial \varepsilon_r^2} = -\frac{2\varepsilon_\infty^2}{\varepsilon_r^3(\varepsilon_\infty + 2)^2}, \quad \frac{\partial^2 F}{\partial \varepsilon_r \partial \varepsilon_\infty} = \frac{4(\varepsilon_\infty - \varepsilon_r^2)}{\varepsilon_r^2(\varepsilon_\infty + 2)^3}. \quad (\text{S28})$$

The $\chi_{\text{H}_2\text{O}}$ -column entry is

$$H_{02} = -\frac{\partial Z^{\text{aq, assoc}}}{\partial \chi_{\text{H}_2\text{O}}} - \frac{\partial Z^{\text{aq, Perm}}}{\partial \chi_{\text{H}_2\text{O}}}, \quad (\text{S29})$$

where $\partial Z^{\text{aq, assoc}} / \partial \chi_{\text{H}_2\text{O}}$ follows from the Wertheim association compressibility [1] through $\chi_{\text{CO}_2}(\chi_{\text{H}_2\text{O}})$, and $\partial Z^{\text{aq, Perm}} / \partial \chi_{\text{H}_2\text{O}}$ is obtained by differentiating the chi cross-derivative system with respect to $\chi_{\text{H}_2\text{O}}$ and propagating through $\mathcal{P}_{ww}$, $\mathcal{P}_{wc}$, g_w , and $V(\partial F / \partial V)$. The Z^{aq} -column entry H_{00} approximates $\partial(Z^{\text{aq, phys}} + Z^{\text{aq, assoc}} + Z^{\text{aq, DH}} + Z^{\text{aq, Perm}}) / \partial Z^{\text{aq}}$ by omitting the implicit Z^{aq} -dependence of $V(\partial \chi_{\text{H}_2\text{O}} / \partial V)$ and $V(\partial \chi_{\text{CO}_2} / \partial V)$ through the chi cross-derivative system. Comparison with centered finite differences confirms this approximation incurs a 1–2% error in H_{00} at typical conditions, which does not impair Newton convergence from warm starts (the other eight entries are exact to machine precision).

S4. Additional Validation Figures

This section supplements the $\text{CO}_2+\text{H}_2\text{O}$ and CO_2+NaCl validation sections of the main text. Figures S1–S4 and S7 provide complete coverage of the $\text{CO}_2+\text{H}_2\text{O}$ and CO_2+NaCl validation datasets at all temperatures and salinities considered. Figures S5–S6 present the extended aqueous density validation against IAPWS-95. The accuracy and efficiency of the simplified flash are further characterized in Table S5 and Figs. S8–S9.

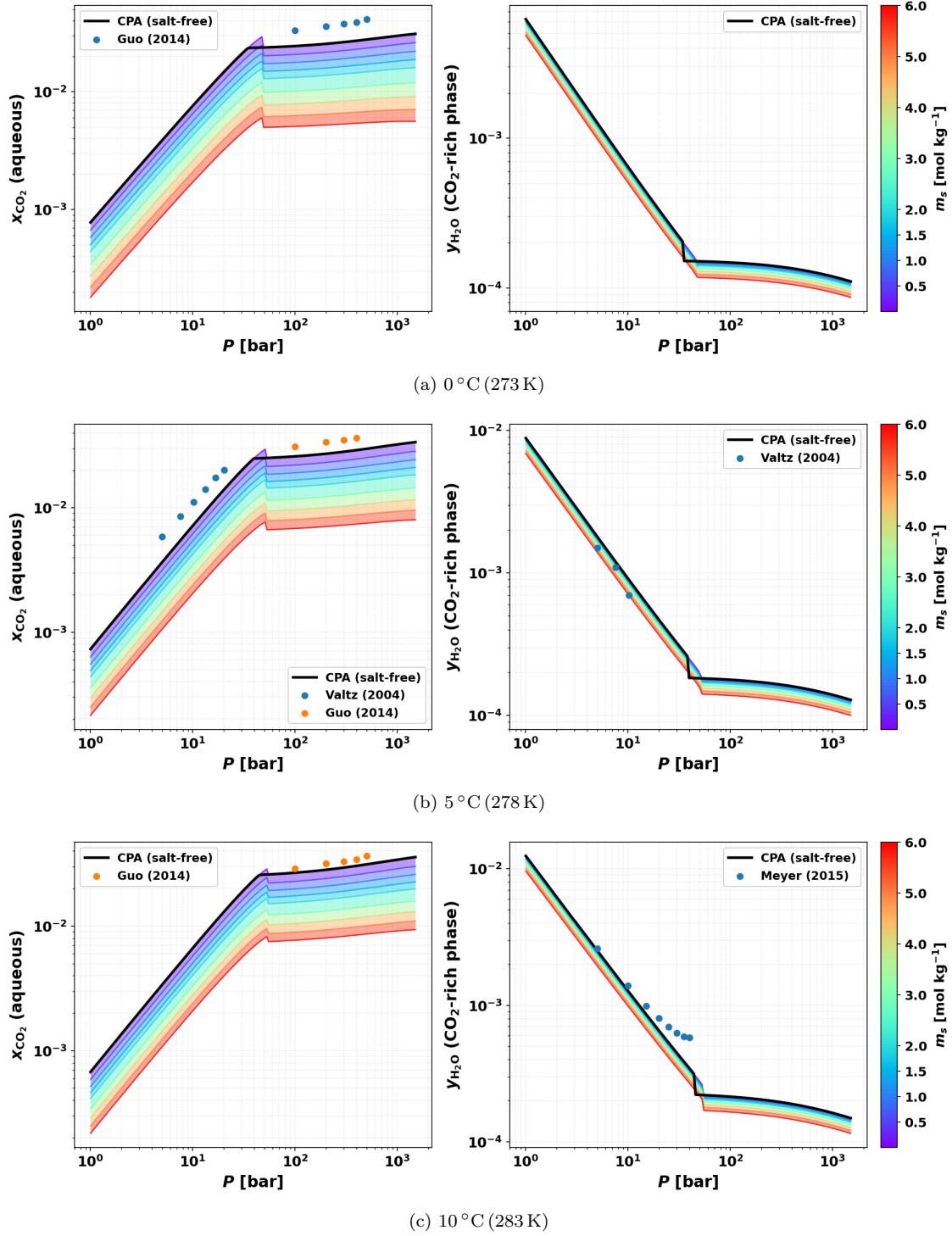
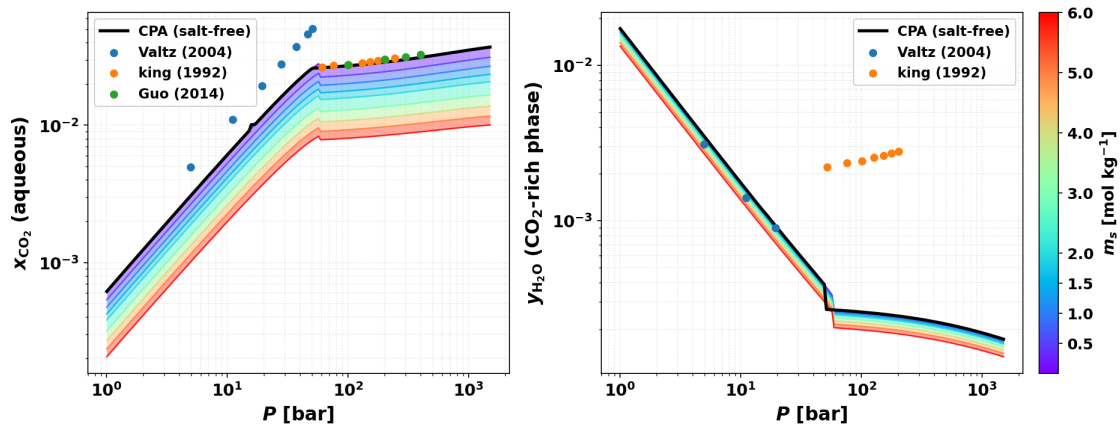
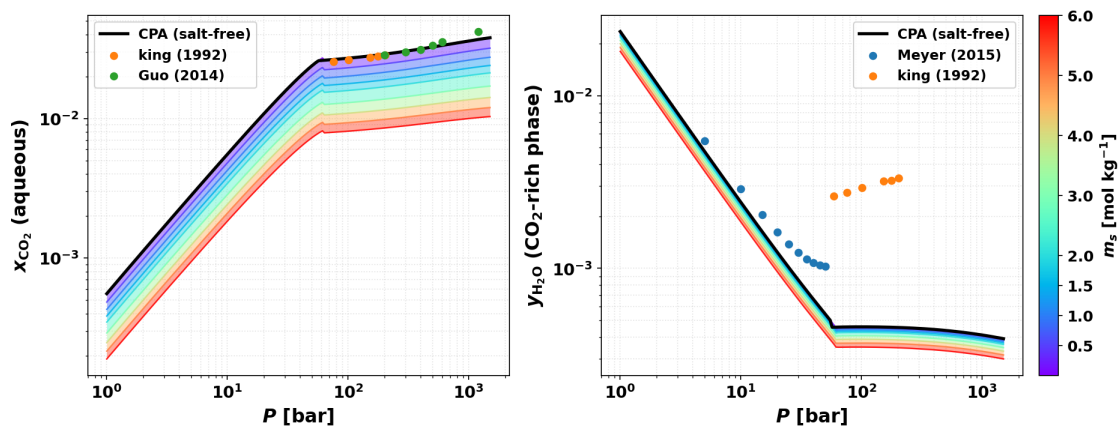


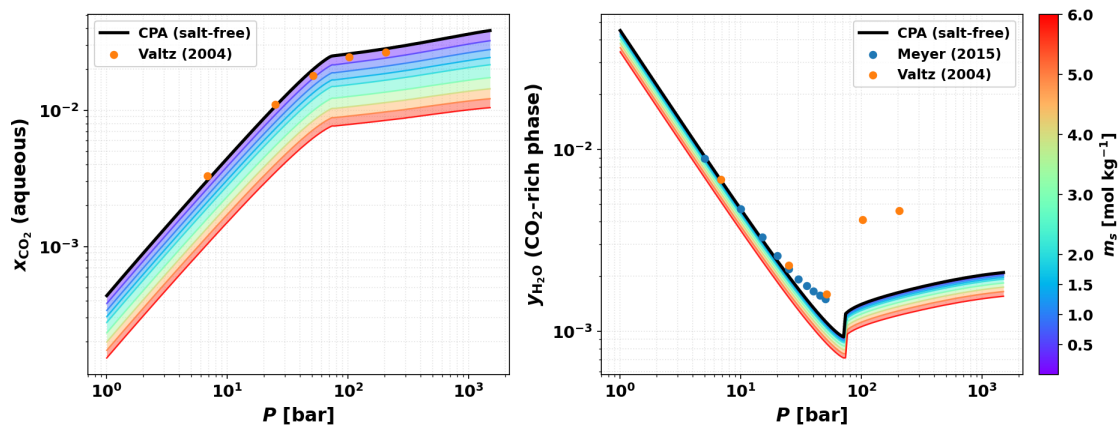
Figure S1: $\text{CO}_2 + \text{H}_2\text{O}$ phase equilibria at all 32 temperatures not shown in Fig. 1 of the main text (0–350 °C), one temperature per row. For each temperature, we show the CO_2 mole fraction in the aqueous phase (x_{CO_2} , left panel) and H_2O mole fraction in the CO_2 -rich phase ($y_{\text{H}_2\text{O}}$, right panel) vs. pressure. Open symbols: experimental data [1]; black solid line: CPA (salt-free); rainbow-colored filled bands: eCPA spanning $m_s = 0\text{--}6 \text{ mol kg}^{-1}$ (violet to red), with discrete molality contours overlaid. This figure continues through the next pages.



(d) 15 °C (288 K)

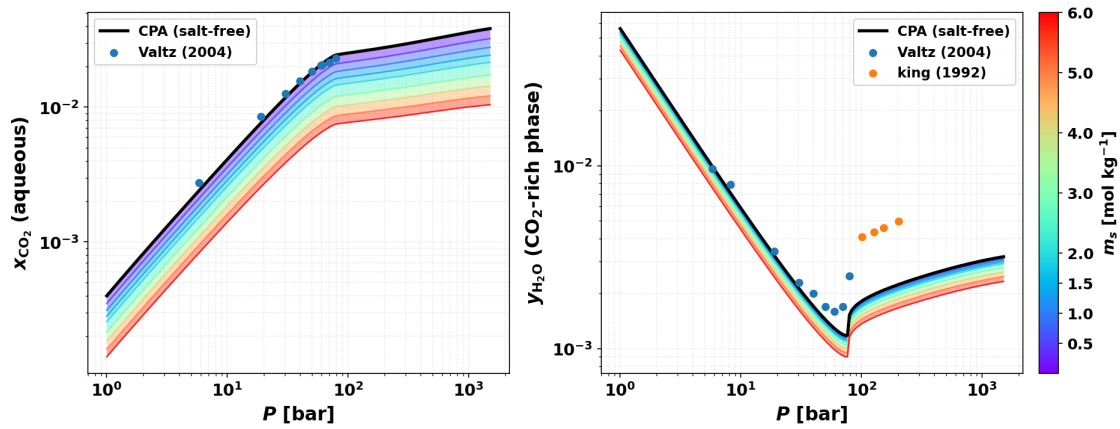


(e) 20 °C (293 K)

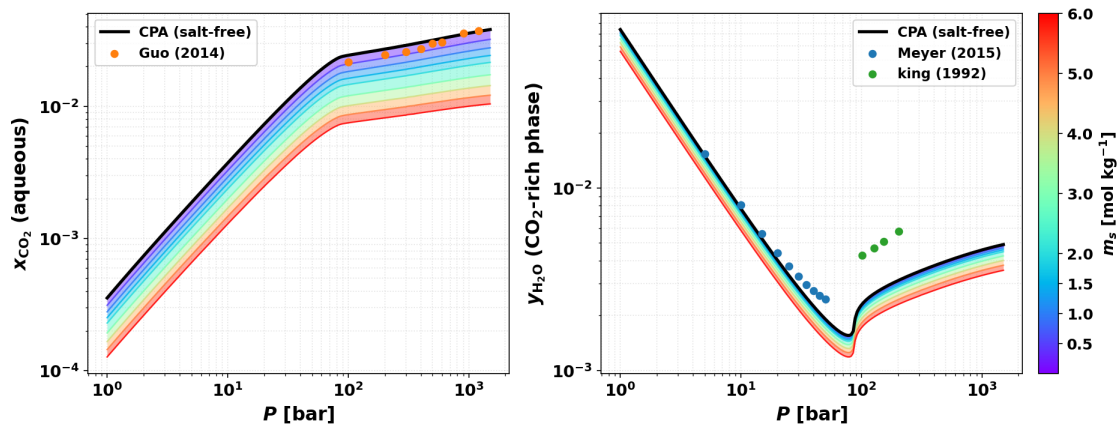


(f) 31 °C (304 K)

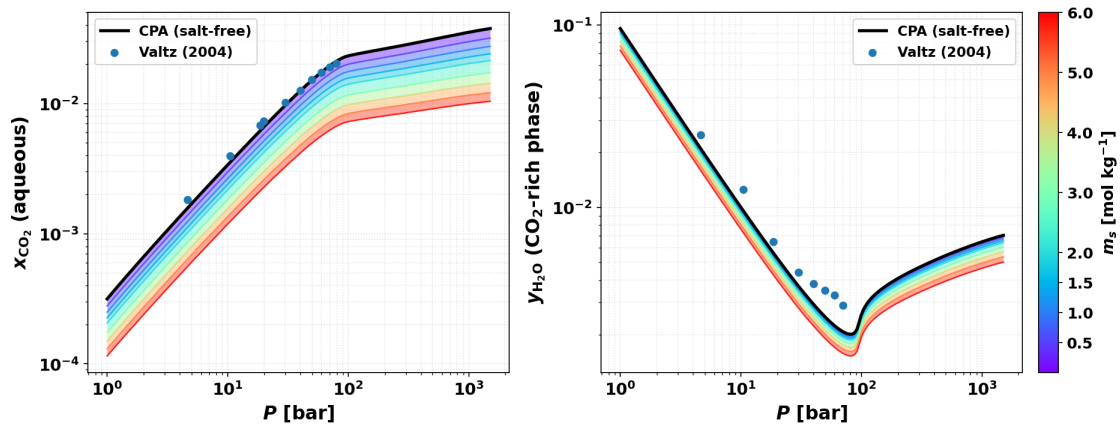
Figure S1: (*continued*)



(g) 35 °C (308 K)

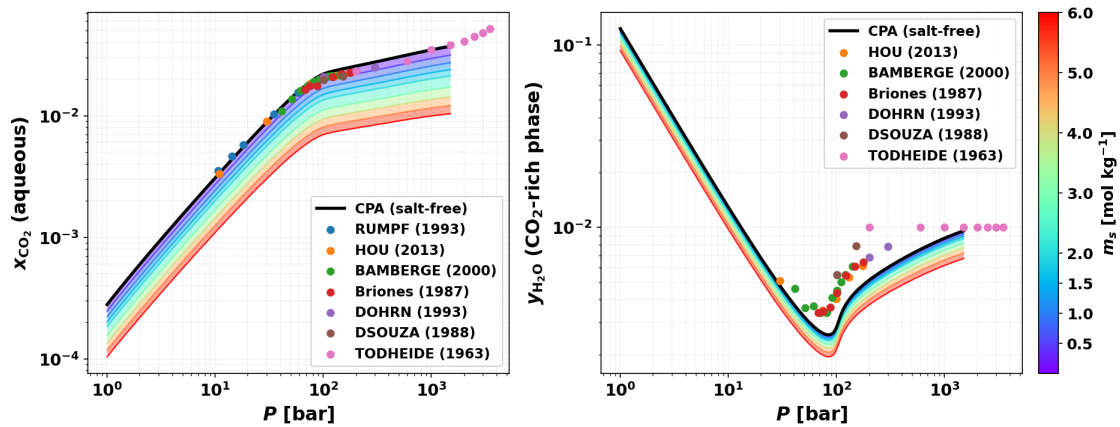


(h) 40 °C (313 K)

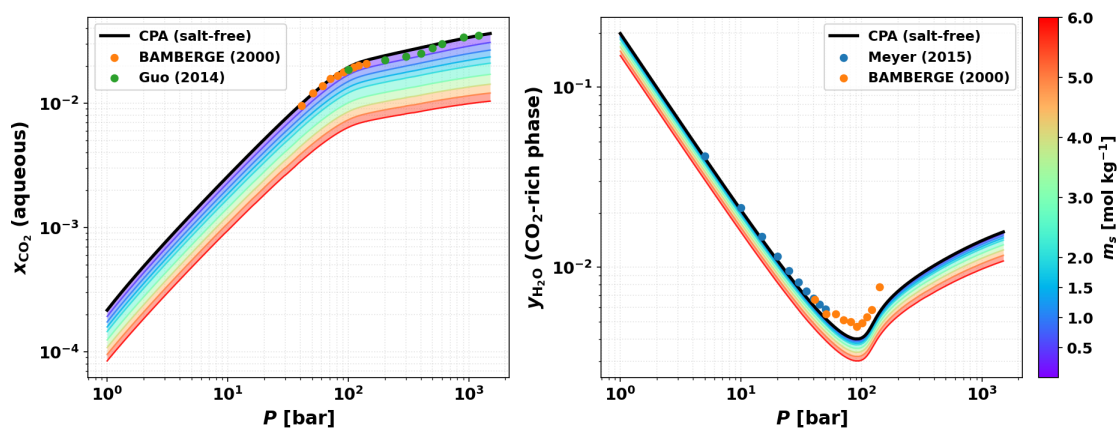


(i) 45 °C (318 K)

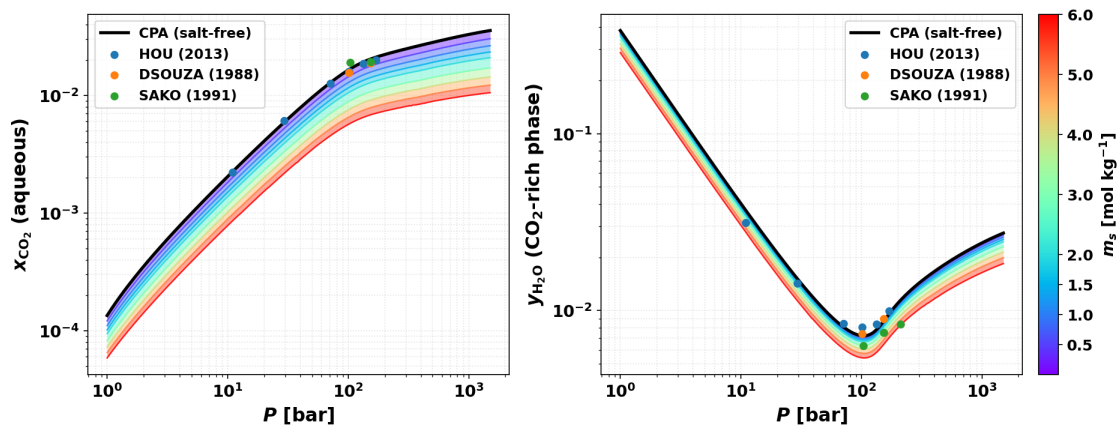
Figure S1: (*continued*)



(j) 50 °C (323 K)

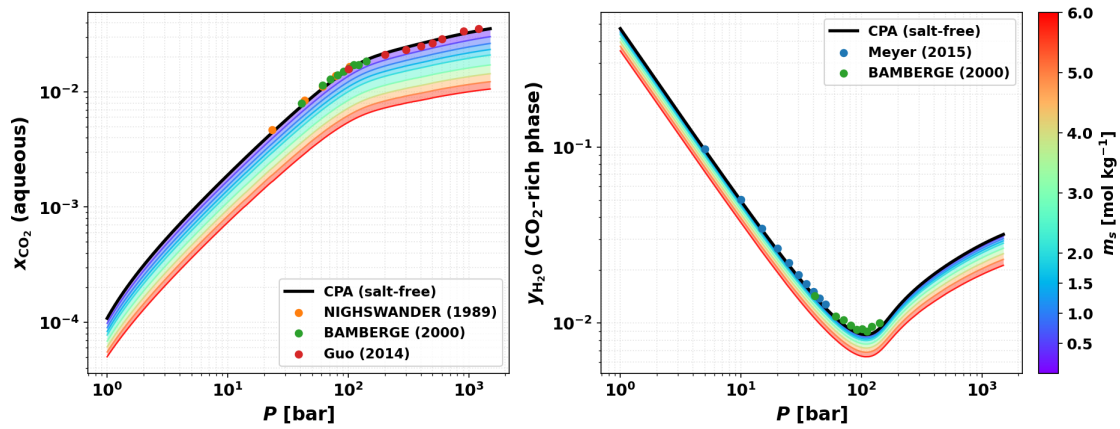


(k) 60 °C (333 K)

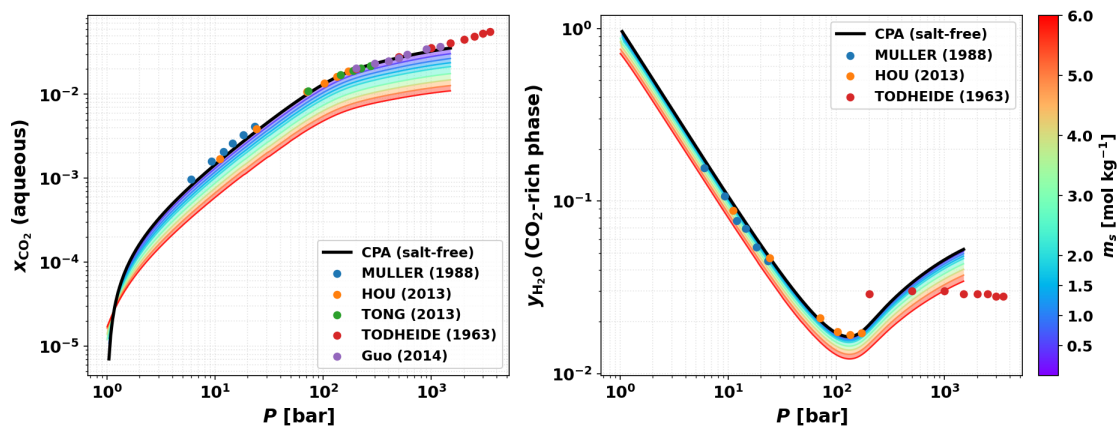


(l) 75 °C (348 K)

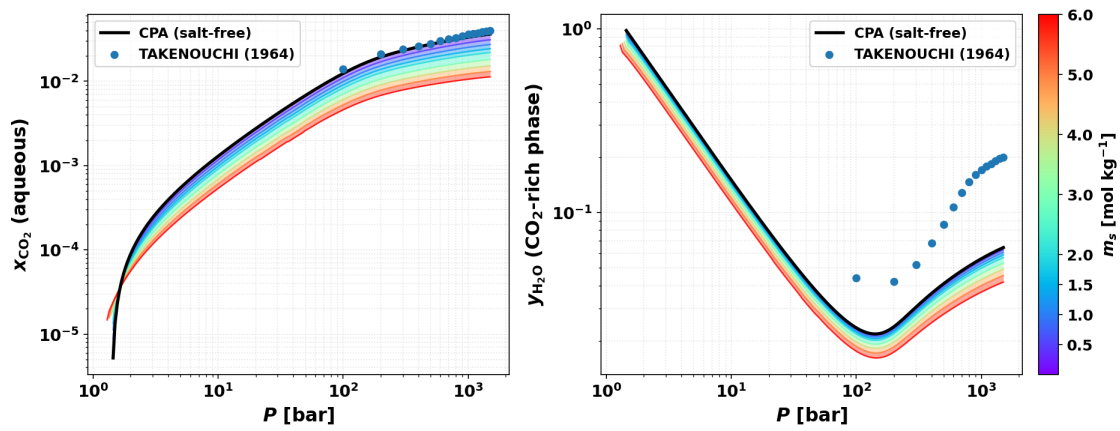
Figure S1: (*continued*)



(m) 80 °C (353 K)

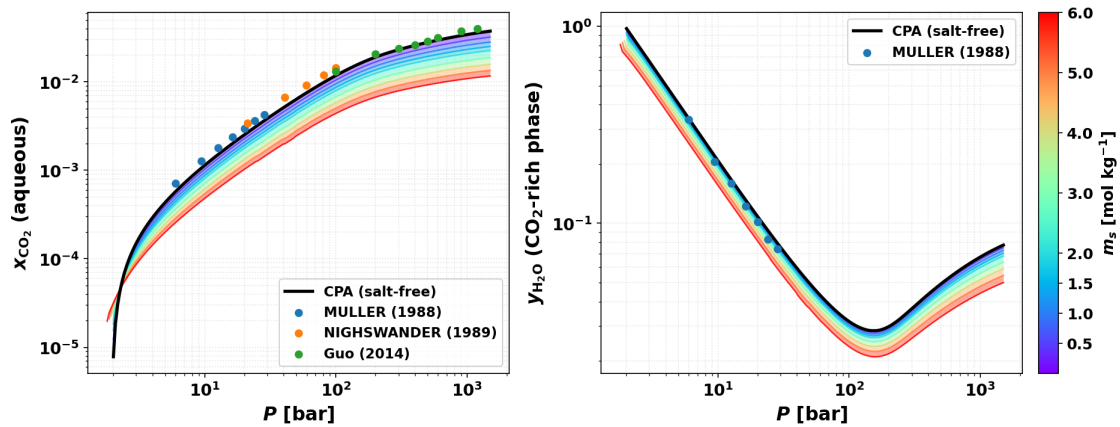


(n) 100 °C (373 K)

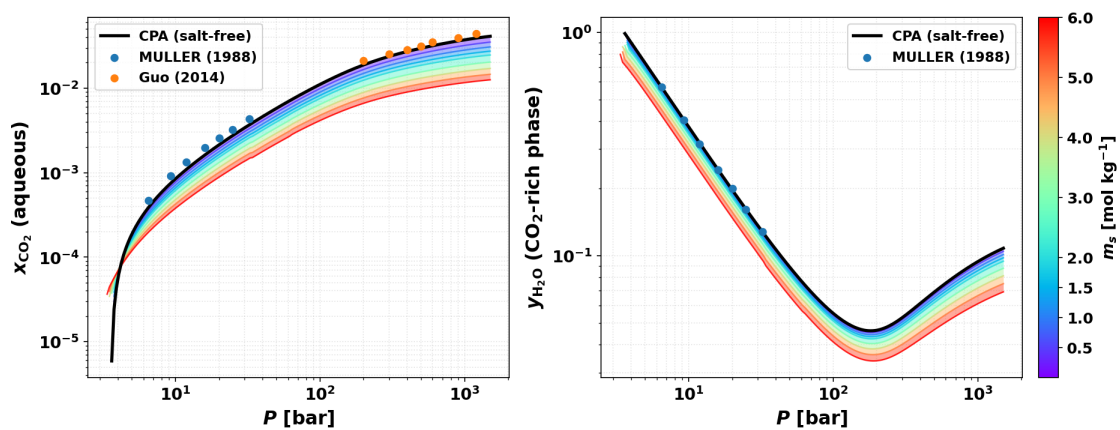


(o) 110 °C (383 K)

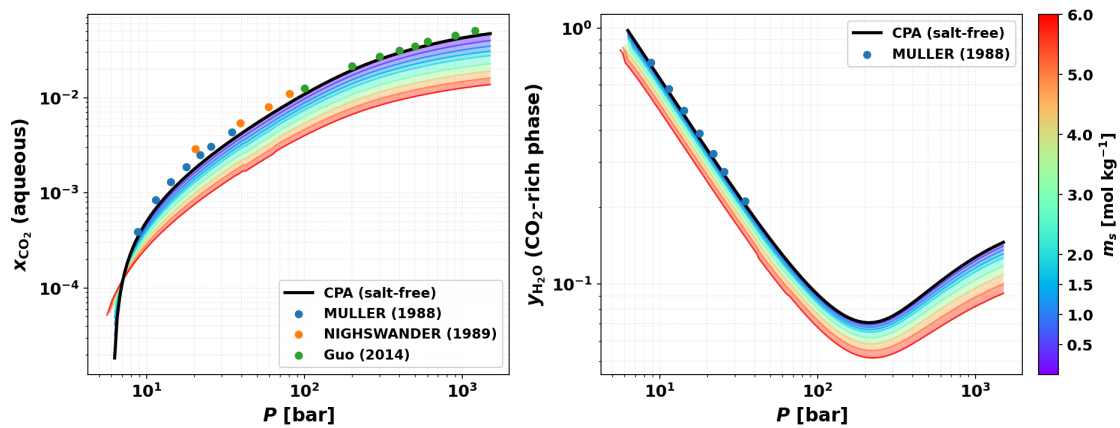
Figure S1: (*continued*)



(p) 120 °C (393 K)

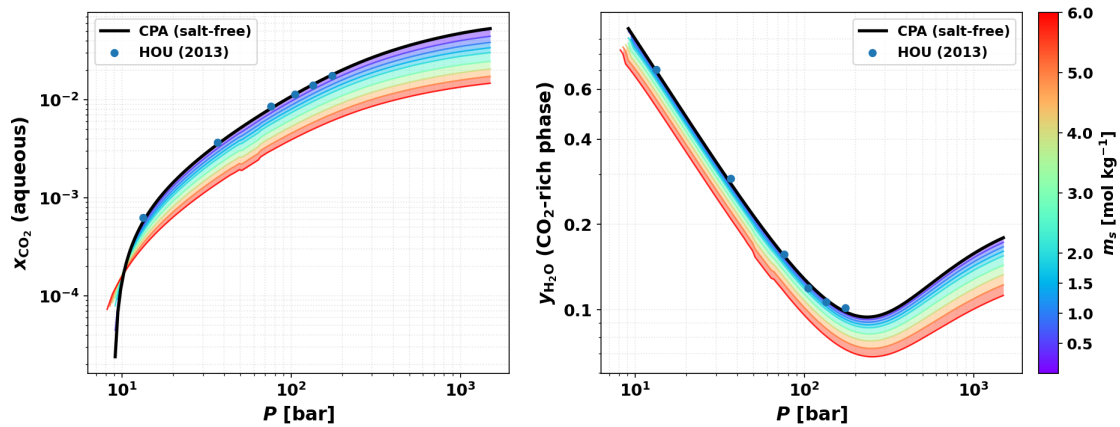


(q) 140 °C (413 K)

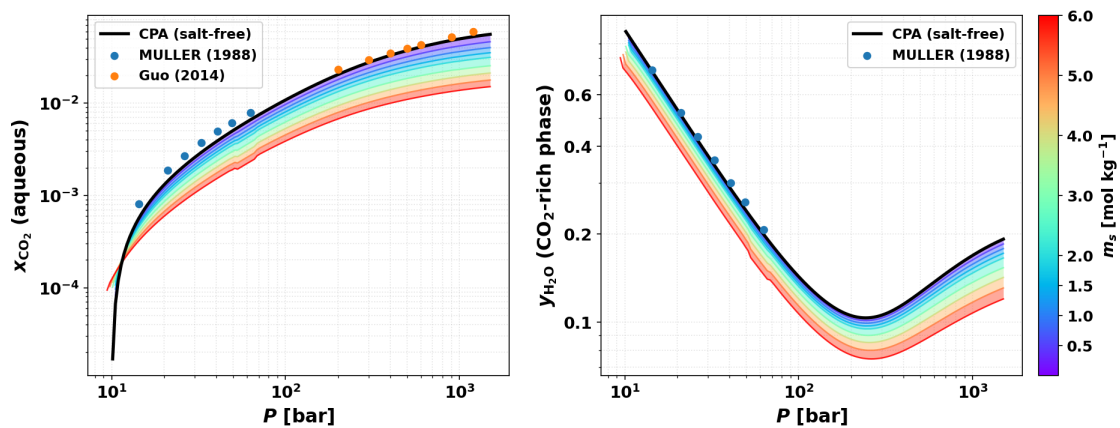


(r) 160 °C (433 K)

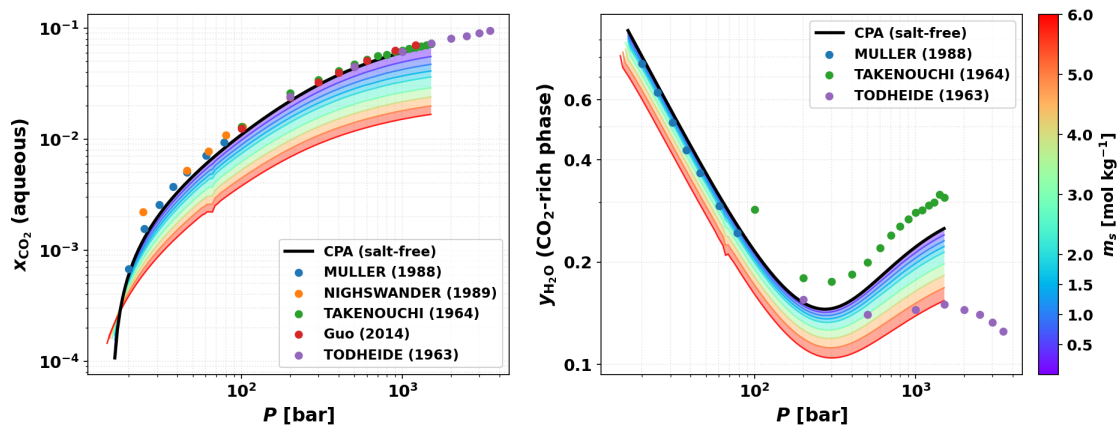
Figure S1: (*continued*)



(s) 175 °C (448 K)

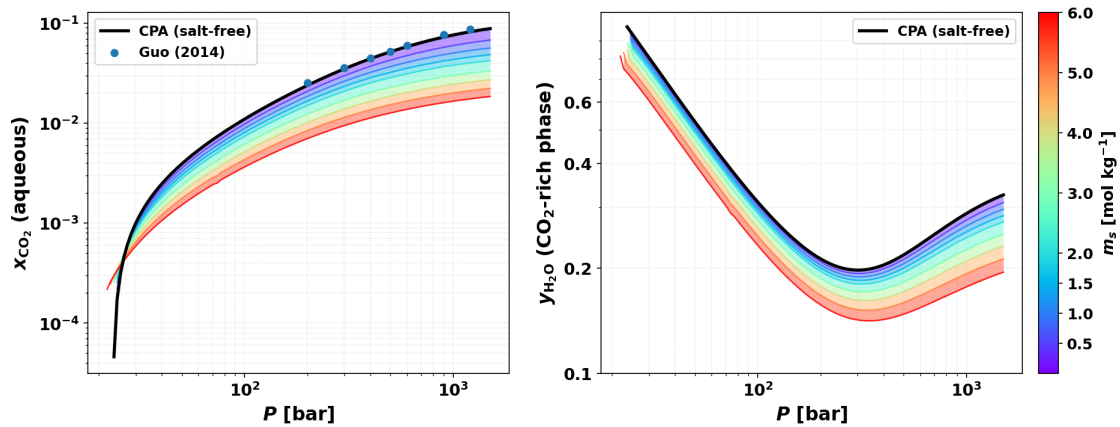


(t) 180 °C (453 K)

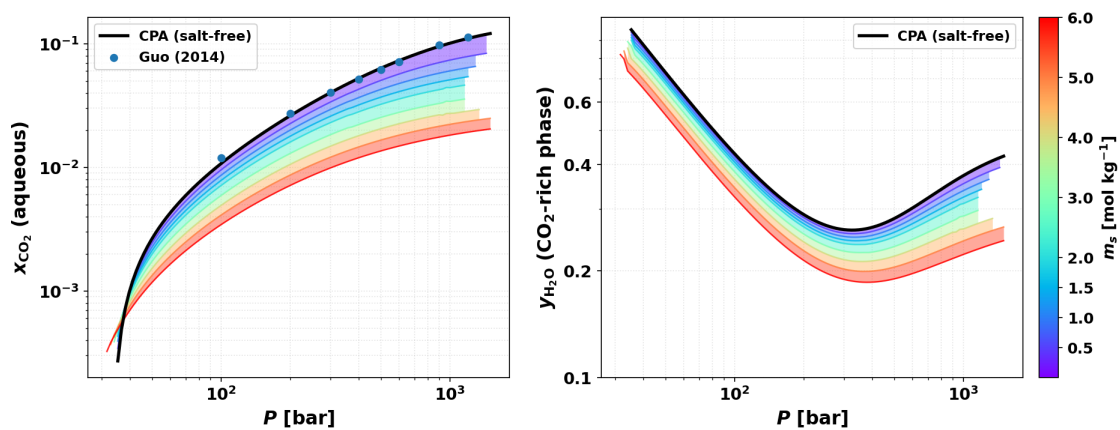


(u) 200 °C (473 K)

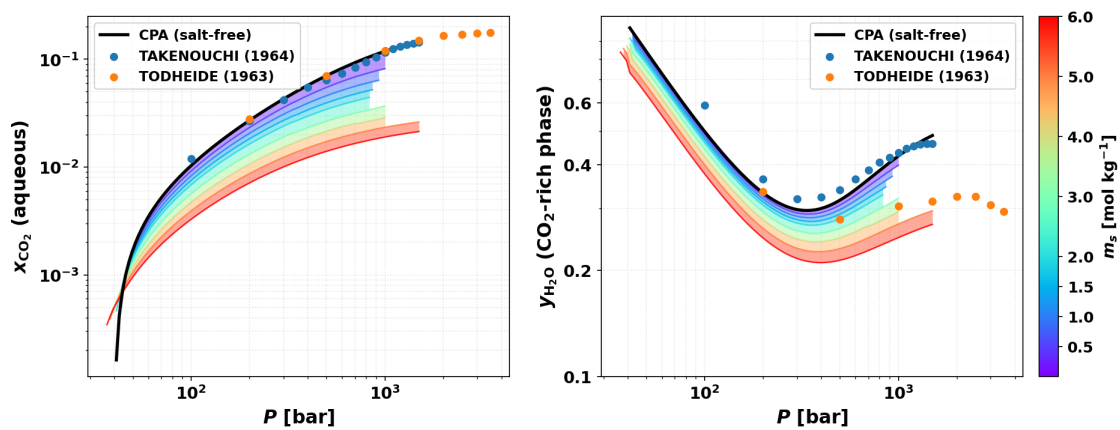
Figure S1: (*continued*)



(v) 220 °C (493 K)

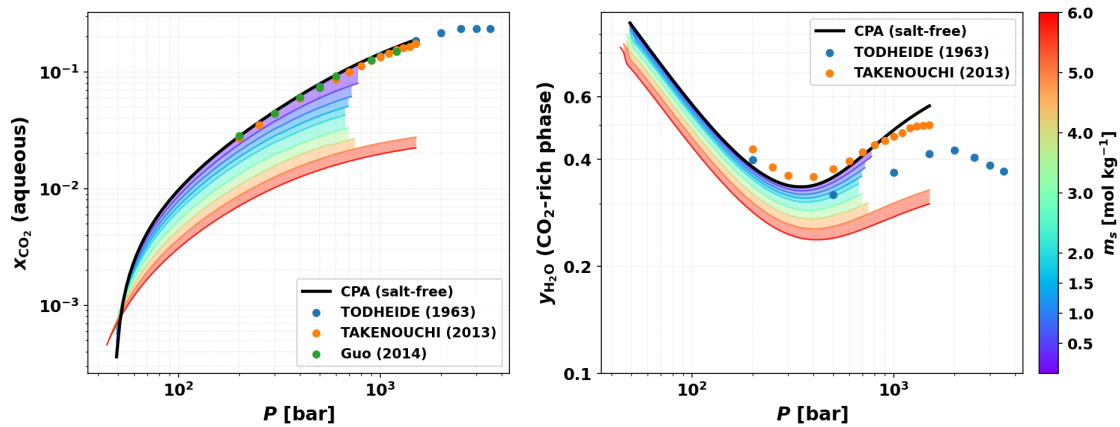


(w) 240 °C (513 K)

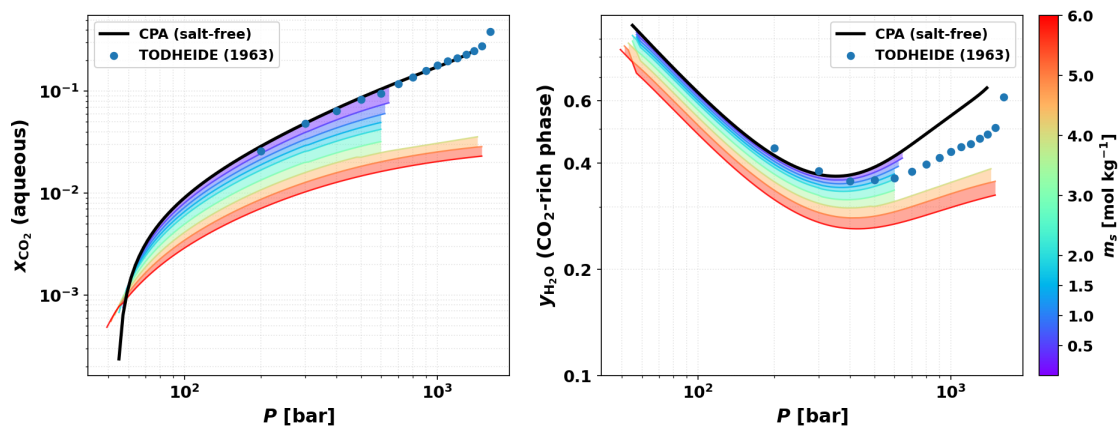


(x) 260 °C (533 K)

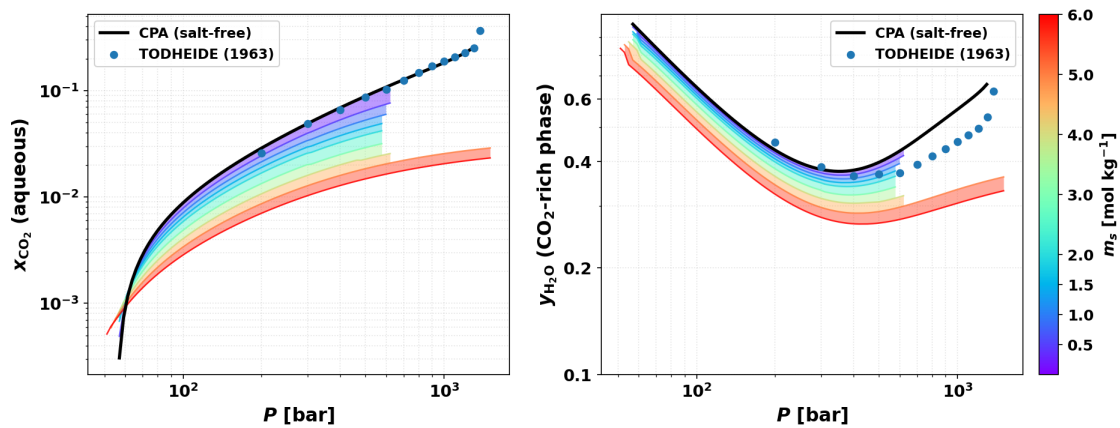
Figure S1: (continued)



(y) 265 °C (538 K)

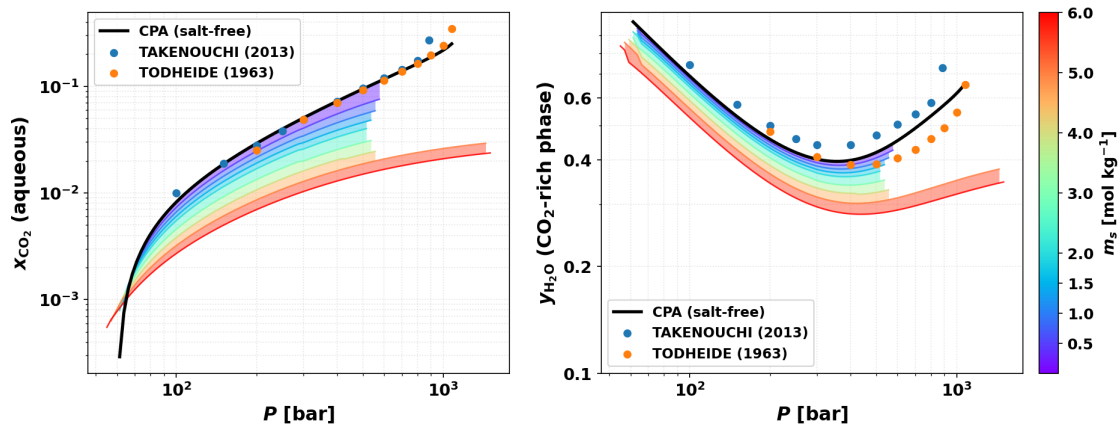


(z) 268 °C (541 K)

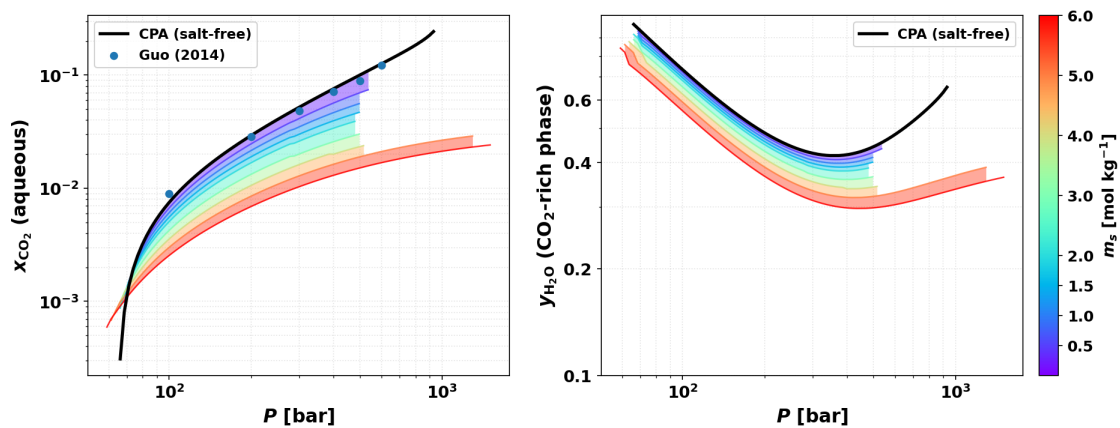


(aa) 270 °C (543 K)

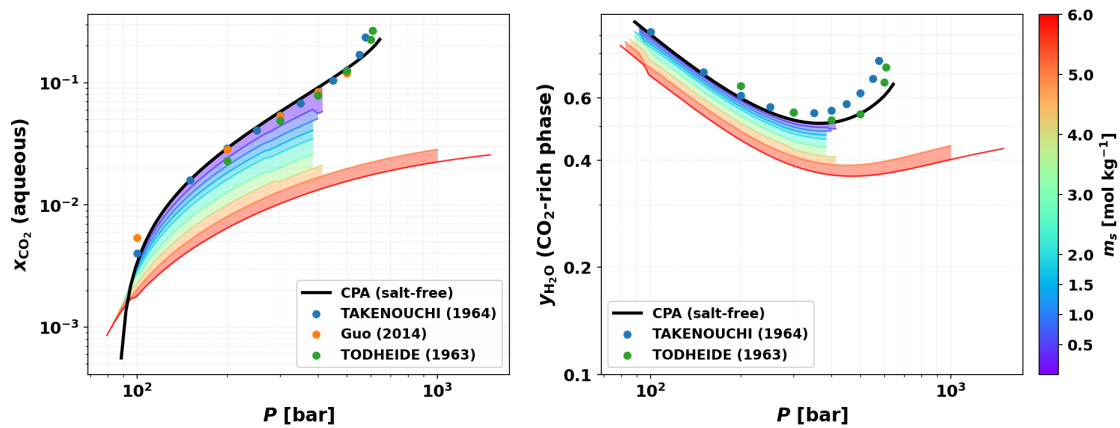
Figure S1: (*continued*)



(ab) 275 °C (548 K)

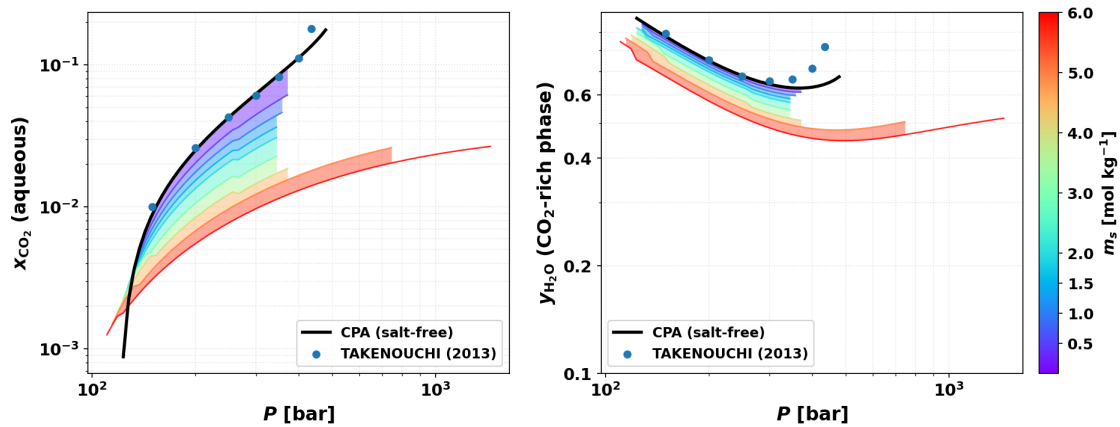


(ac) 280 °C (553 K)

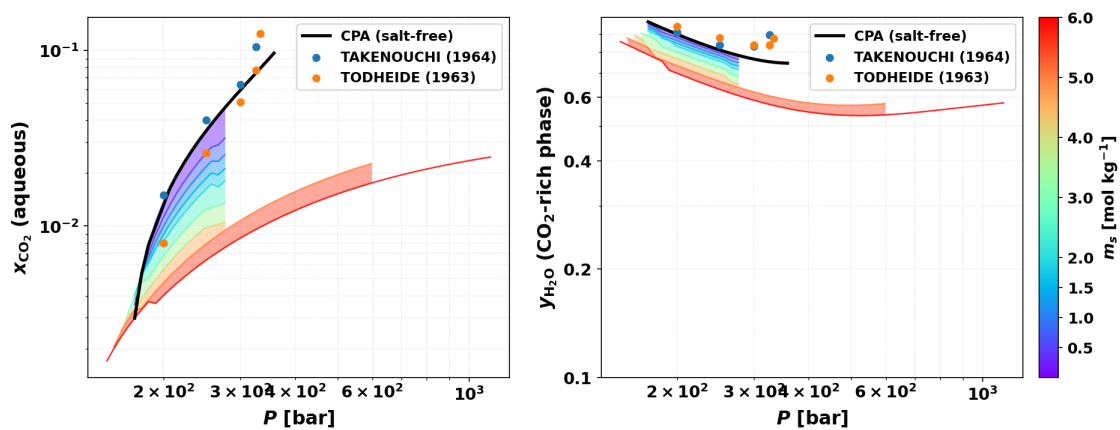


(ad) 300 °C (573 K)

Figure S1: (*continued*)



(ae) 325 °C (598 K)



(af) 350 °C (623 K)

Figure S1: (*continued*)

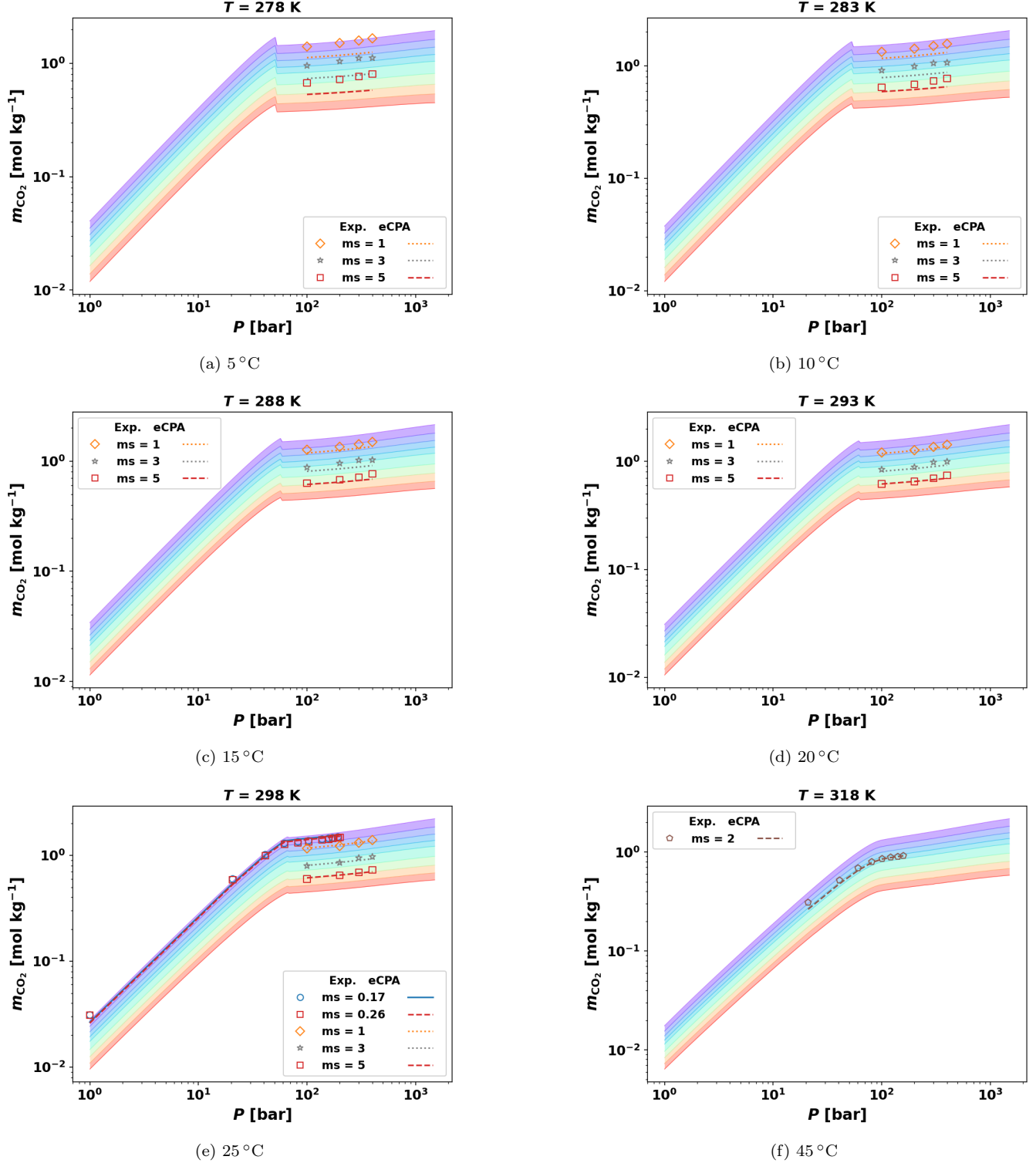
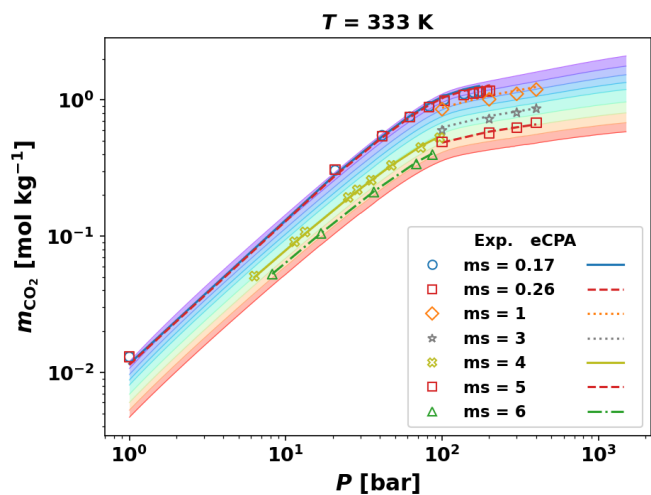
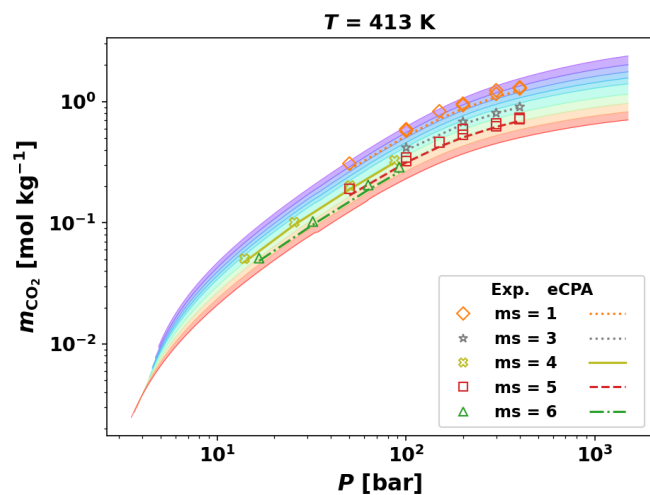


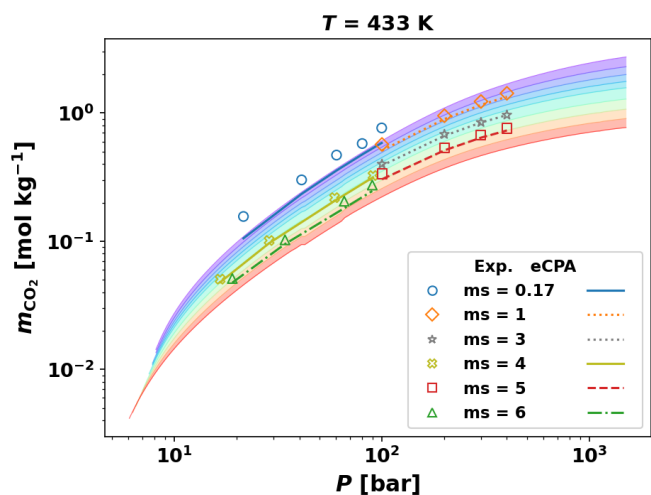
Figure S2: CO₂ solubility in NaCl brine at thirteen temperatures not shown in Fig. 2 of the main text (5–350 °C), including data at high NaCl molalities up to $m_s = 6 \text{ mol kg}^{-1}$ from multiple sources. Each panel shows CO₂ molality m_{CO_2} [mol kg⁻¹] vs. pressure (log-log). Open symbols: experimental data [2, 3, 4, 5, 6, 7, 8, 9, 10]; discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions across $m_s = 0\text{--}6 \text{ mol kg}^{-1}$ (color scale as in Fig. 1 of the main text). At $T \geq 400 \text{ °C}$, eCPA predicts single-phase (fully miscible) behavior for all available experimental pressures.



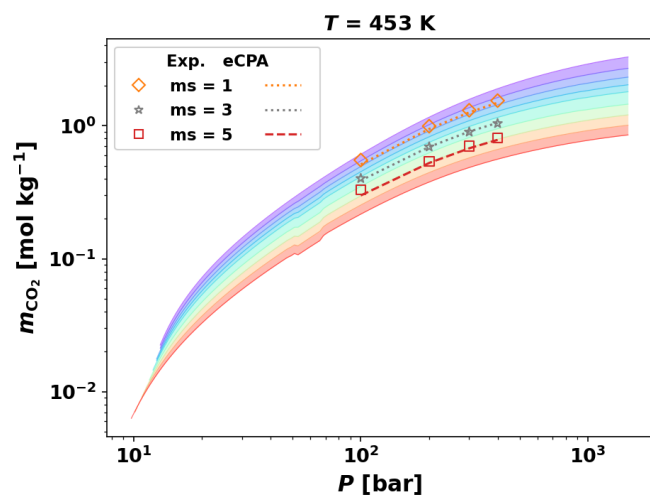
(g) 60 °C



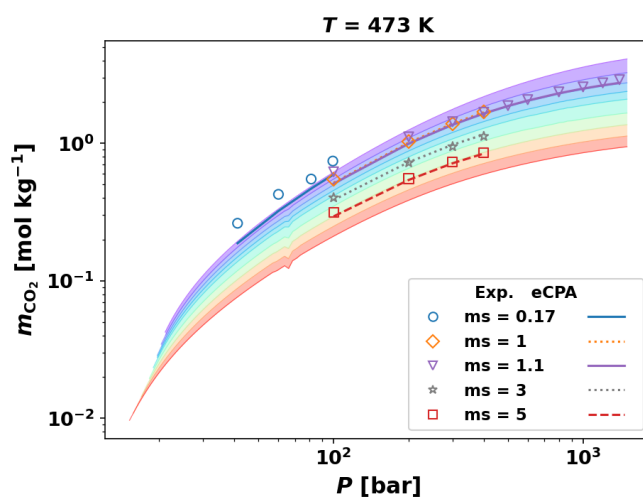
(h) 140 °C



(i) 160 °C



(j) 180 °C



(k) 200 °C

Figure S2: (*continued*)

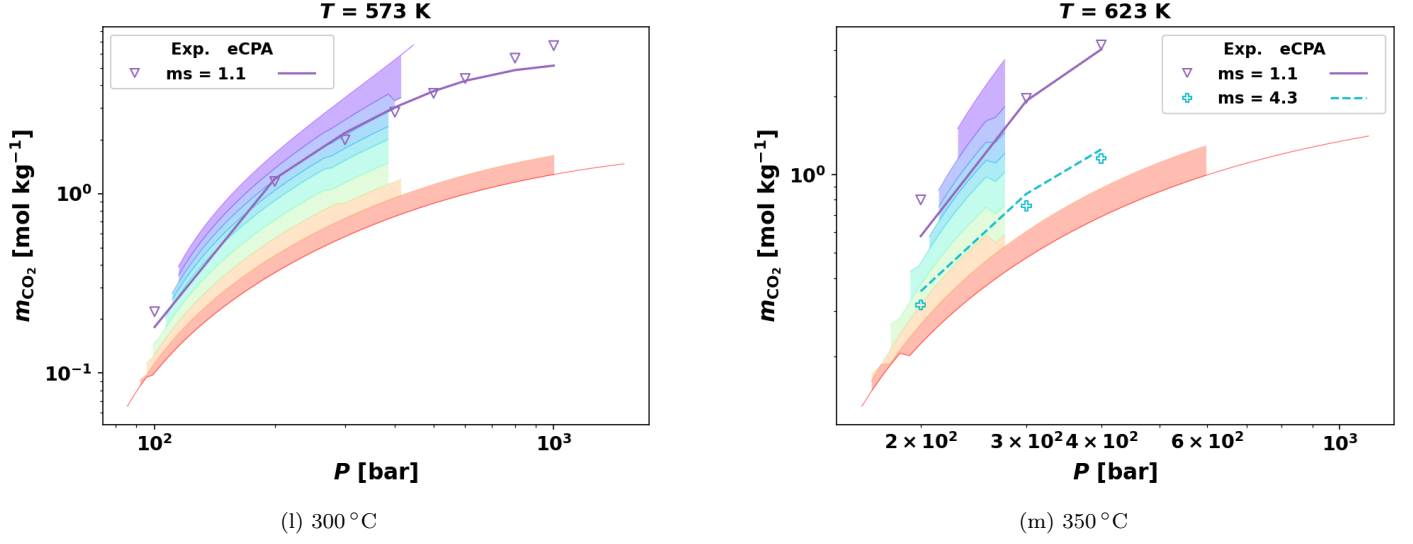


Figure S2: (*continued*)

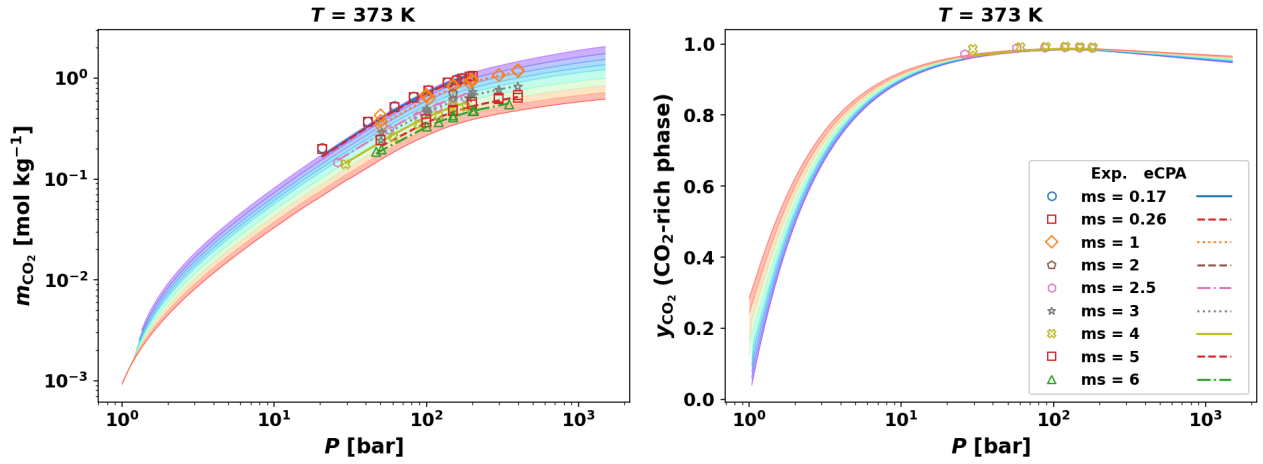


Figure S3: CO₂ + NaCl brine phase equilibria at $T = 100$ °C for which both CO₂ molality m_{CO_2} (left panel) and CO₂ mole fraction in the CO₂-rich phase y_{CO_2} (right panel) data are available. Open symbols: experimental data [4, 5, 2, 11, 12, 6, 13, 14, 9, 10, 15]; discrete lines with markers: eCPA at the experimental NaCl molalities. Rainbow-colored background bands: continuous eCPA predictions across $m_s = 0$ –6 mol kg⁻¹ (color scale as in Fig. 1 of the main text).

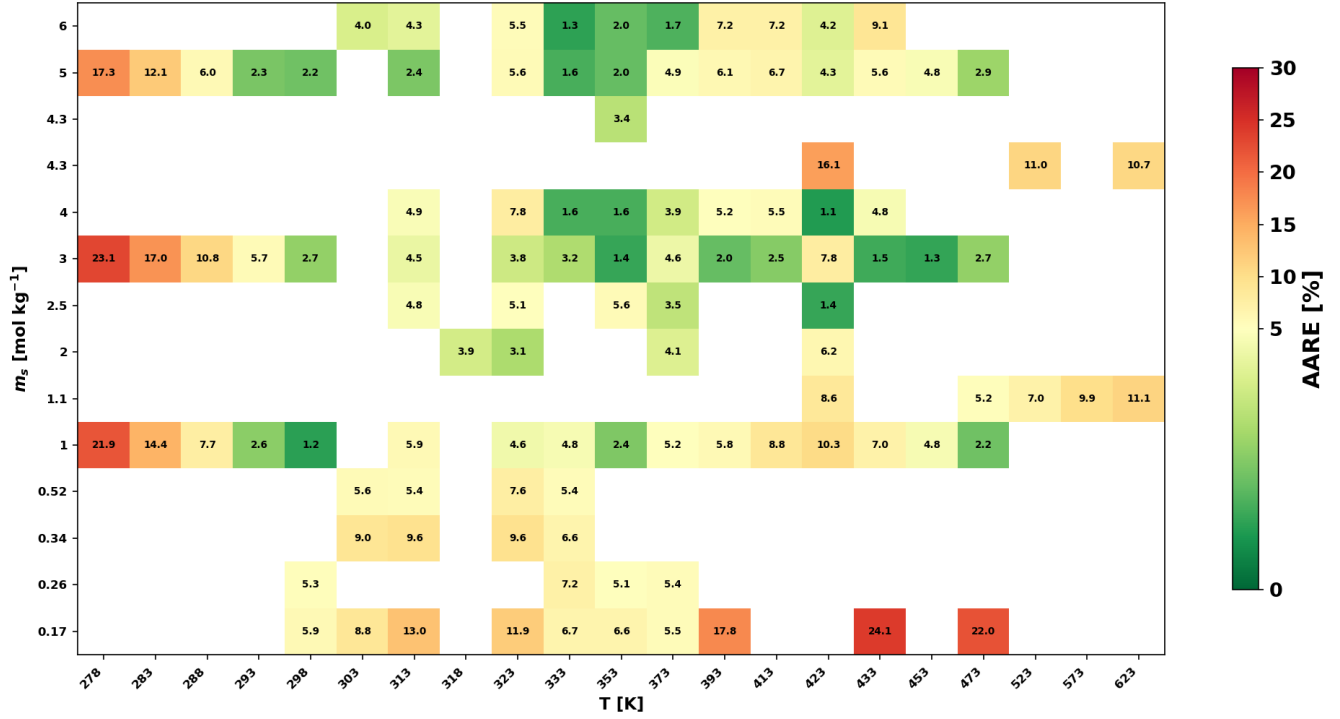


Figure S4: Heatmap of absolute relative error (AARE, %) for the $\text{CO}_2 + \text{NaCl}$ validation dataset ($T = 5\text{--}350^\circ\text{C}$), showing all individual NaCl molalities present in the experimental data. Color scale is diverging at 5%: green $\leq 5\%$, red $\geq 10\%$. Empty cells indicate no experimental data at that (T, m_s) combination.

S4.1. Aqueous-phase density: extended validation

We validate pure-water CPA/eCPA density (with the temperature-dependent P  neloux shift, Section S2) against the IAPWS-95 international standard [16], whose documented uncertainty in liquid-water density is $\pm 0.001\text{--}0.02\%$ [16] — well below the eCPA model error — over a systematic grid of 26 temperatures ($T = 0\text{--}350^\circ\text{C}$) and 20 pressures ($P = 1\text{--}2\,000$ bar), restricted to liquid/compressed-water conditions (475 points; vapour excluded). The eCPA density is $\rho = M_{\text{H}_2\text{O}} / (Z^{\text{aq}}RT/P + c_{\text{H}_2\text{O}}(T))$, where $c_{\text{H}_2\text{O}}(T)$ is the polynomial from Section S2.

The parity plot (Fig. S5) shows tight agreement across the full range. The signed error map (Fig. S6) is nearly flat, with residual errors $< 0.7\%$ at most conditions; the largest deviations ($\approx 3\%$) occur near 350°C , close to the critical point of water, where cubic-EoS predictions are least accurate. Overall: AARE = 0.30% , bias = -0.20% , maximum ARE = 3.0% (475 liquid conditions). For subsurface CO_2 storage conditions ($T \leq 120^\circ\text{C}$, $P = 75\text{--}600$ bar, $N = 136$), the AARE is 0.12% .

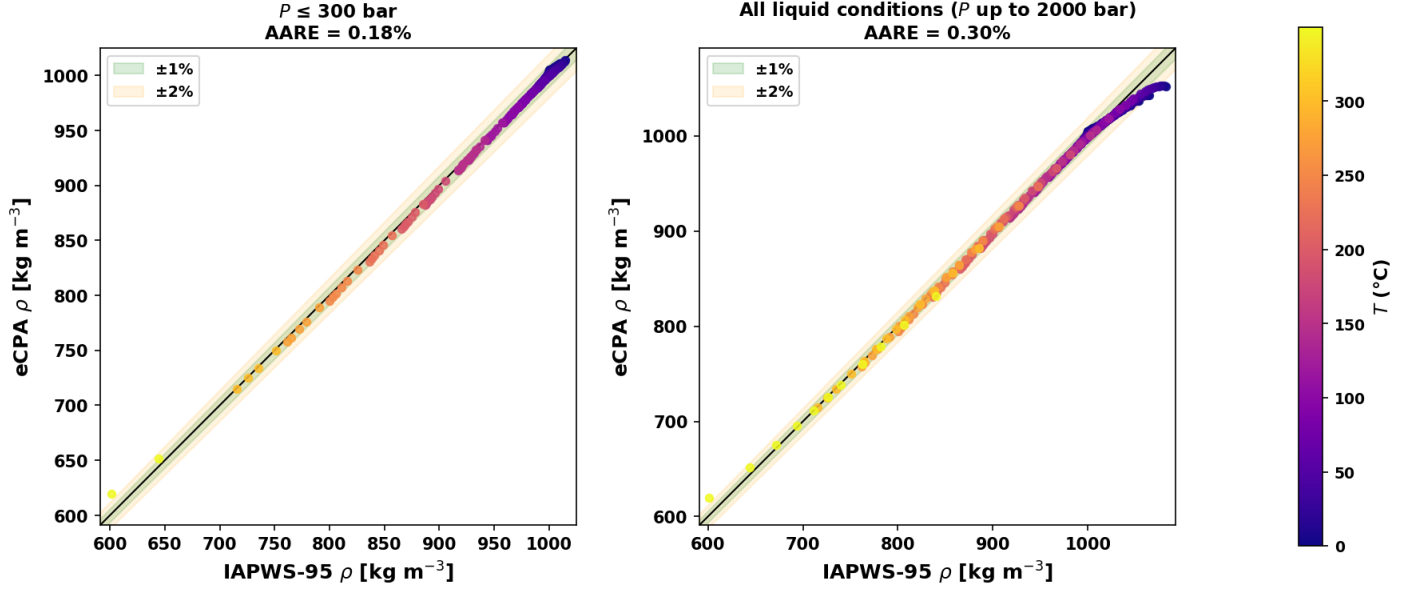


Figure S5: Parity plot: eCPA pure-water density versus IAPWS-95 [16] (475 liquid/compressed-water conditions, $T = 0\text{--}350\text{ }^{\circ}\text{C}$, $P = 1\text{--}2000\text{ bar}$). Left panel: $P \leq 300\text{ bar}$; right panel: all conditions up to 2000 bar. Points are colored by temperature; shaded bands indicate $\pm 1\%$ and $\pm 2\%$ deviations. The temperature-dependent Péneloux shift (Section S2) is applied.

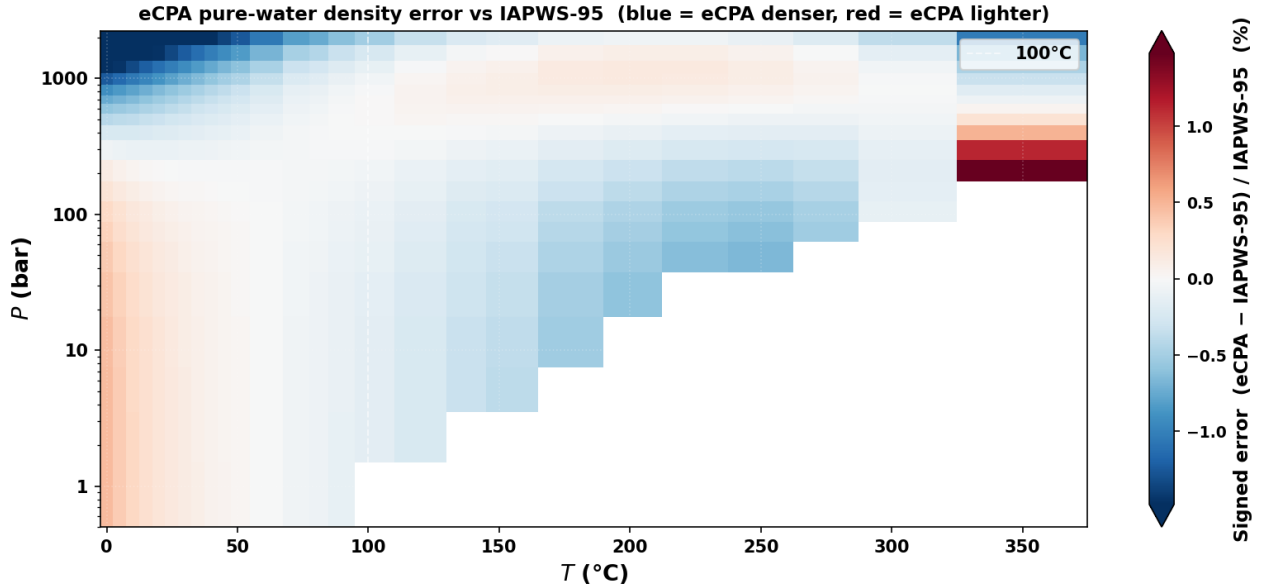


Figure S6: Signed relative error $(\text{eCPA} - \text{IAPWS-95}) / \text{IAPWS-95} (\%)$ of pure-water density predictions over $T = 0\text{--}350\text{ }^{\circ}\text{C}$ and $P = 1\text{--}2000\text{ bar}$ (475 liquid/compressed-water conditions; vapour excluded; P -axis logarithmic). The temperature-dependent Péneloux shift (Section S2) is applied. The error field is nearly flat ($< 0.7\%$ over most of the domain); the largest deviations ($\approx 3\%$) occur near $350\text{ }^{\circ}\text{C}$ at moderate pressures, close to the critical point of water.

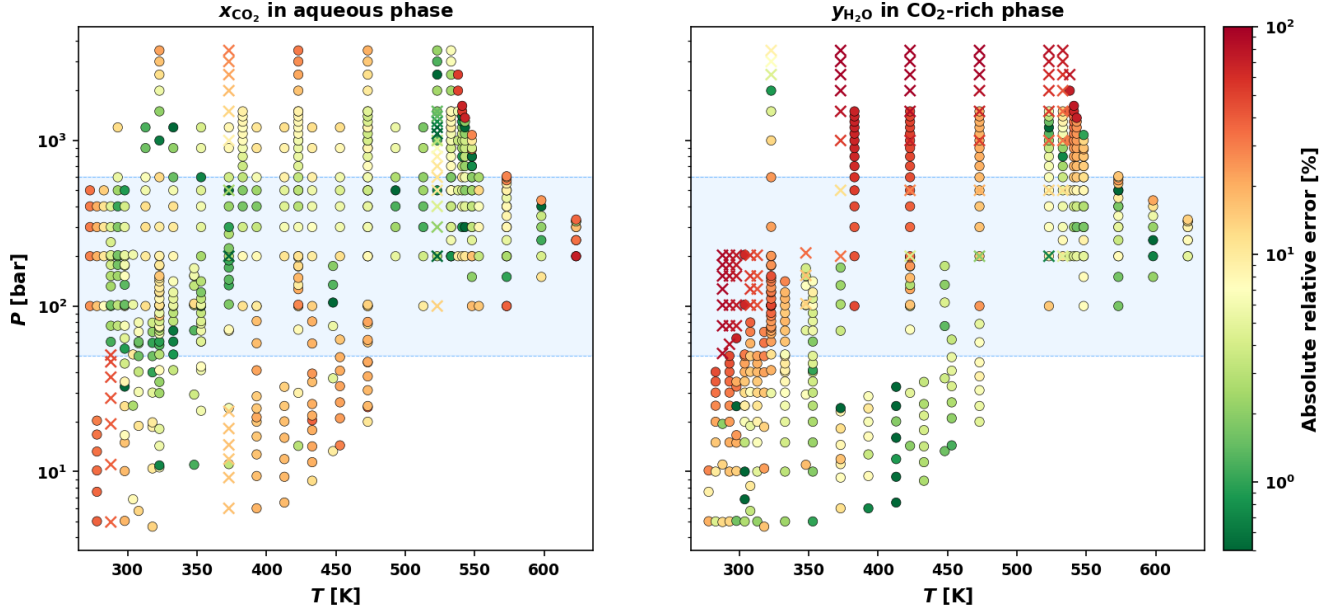


Figure S7: Absolute relative error of eCPA predictions for the $\text{CO}_2 + \text{H}_2\text{O}$ binary as a function of temperature and pressure. Left: CO_2 solubility in the aqueous phase (x_{CO_2}). Right: H_2O content in the CO_2 -rich phase ($y_{\text{H}_2\text{O}}$). Marker color indicates error magnitude (green $< 5\%$, yellow $\approx 10\%$, red $> 30\%$); crosses mark points flagged as outliers by cross-reference consistency analysis. The blue shaded band highlights the pressure range most relevant to subsurface CO_2 storage ($P = 50\text{--}600$ bar).

S4.2. Simplified flash: detailed accuracy and efficiency

Table S5 reports the AARE in dissolved CO_2 molality m_{CO_2} and the mean $y_{\text{H}_2\text{O}}$ in the CO_2 -rich phase from the full flash as a function of temperature, evaluated over 3,285 two-phase conditions spanning $P = 25\text{--}800$ bar, $z_{\text{CO}_2} = 0.01\text{--}0.70$, and $m_s = 1\text{--}6$ mol kg^{-1} . The two quantities track each other closely: the simplified flash introduces an error in m_{CO_2} that equals, to a good approximation, the water content it neglects.

Table S5: Accuracy of the simplified flash ($y_{\text{H}_2\text{O}} = 0$) relative to the full K-value SSI flash, evaluated over 3,285 two-phase conditions spanning $P = 25\text{--}800$ bar, $z_{\text{CO}_2} = 0.01\text{--}0.70$, and $m_s = 1\text{--}6$ mol kg⁻¹. AARE: average absolute relative error in CO₂ molality m_{CO_2} ; $\langle y_{\text{H}_2\text{O}} \rangle$: mean H₂O mole fraction in the CO₂-rich phase from the full flash (proxy for the assumption error).

T (K)	T (°C)	N	AARE in m_{CO_2} (%)	$\langle y_{\text{H}_2\text{O}} \rangle$ (mol%)
283	10	239	0.03	0.02
298	25	239	0.11	0.08
313	40	240	0.37	0.27
323	50	240	0.62	0.45
333	60	240	0.93	0.68
348	75	239	1.55	1.15
353	80	240	1.80	1.34
373	100	238	3.16	2.35
393	120	239	5.37	3.91
413	140	237	8.95	6.19
423	150	237	11.67	7.72
453	180	235	25.49	14.01
473	200	231	45.85	19.79
523	250	191	121.28	33.51

Figure S8 summarises the AARE in m_{CO_2} , the mean $y_{\text{H}_2\text{O}}$ from the full flash, and the wall-time speedup as functions of temperature. Figure S9 breaks the AARE down by NaCl molality; the error curves for all salinities are nearly coincident, confirming that the accuracy of the $y_{\text{H}_2\text{O}} = 0$ approximation is controlled entirely by temperature.

The mean wall time is essentially identical to the full flash (speedup 0.8 $\times$): Brent root-finding requires on average 11.3 inner Newton evaluations, whereas the full SSI needs 6.4 outer iterations each involving at most two inner solves. The inner Newton cost per call is similar in both paths, so the Brent overhead offsets the gain from eliminating the outer salt iteration. The principal benefits of the simplified flash are therefore conceptual — the equilibrium reduces to a single scalar CO₂-solubility equation — and robustness in salt-free settings where the ternary outer salt loop is absent by construction.

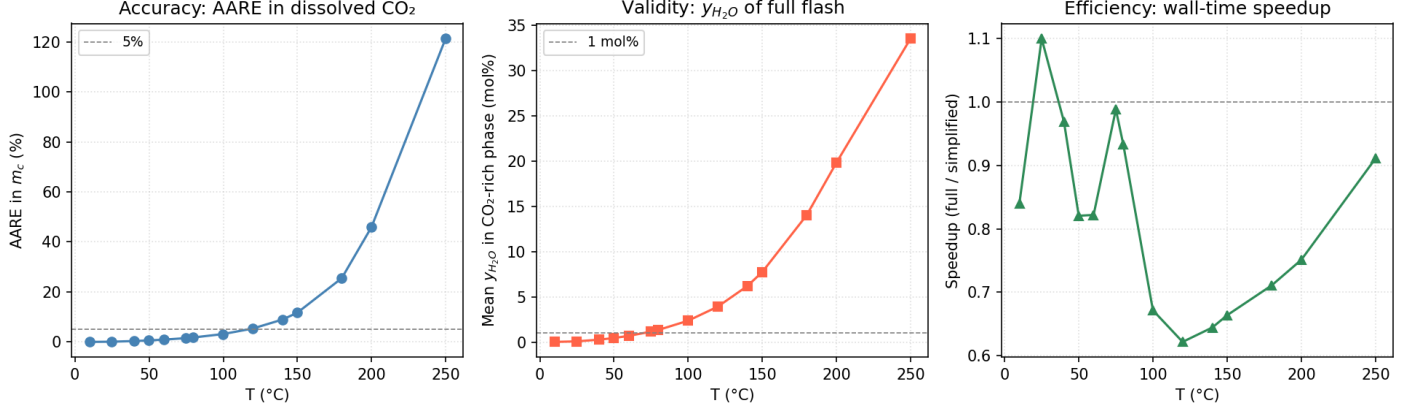


Figure S8: Simplified flash benchmark over 3,285 two-phase conditions ($P = 25\text{--}800$ bar; $z_{\text{CO}_2} = 0.01\text{--}0.70$; $m_s = 1\text{--}6$ mol kg $^{-1}$). **Left:** AARE in dissolved CO_2 molality m_{CO_2} as a function of temperature; the dashed line marks 5%. **Middle:** mean H_2O mole fraction in the CO_2 -rich phase from the full flash, which directly quantifies the magnitude of the $y_{\text{H}_2\text{O}} = 0$ assumption; dashed line at 1 mol%. **Right:** ratio of mean full-flash wall time to simplified-flash wall time; the two methods are essentially iso-cost across all temperatures.

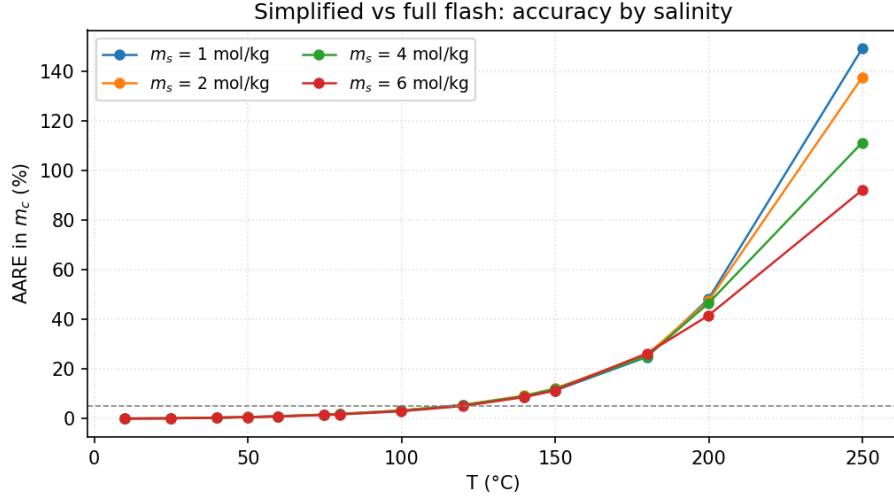


Figure S9: AARE in dissolved CO_2 molality m_{CO_2} of the simplified flash versus the full K-value SSI, broken down by NaCl molality ($m_s = 1\text{--}6$ mol kg $^{-1}$). The error curves for different salinities are nearly coincident, confirming that the accuracy of the $y_{\text{H}_2\text{O}} = 0$ approximation is controlled entirely by temperature and is insensitive to brine salinity. The dashed line marks 5%.

S5. Computational Performance Figures

This section supplements the convergence and efficiency analysis of the main text. Figure S10 shows the SSI+Newton iteration breakdown and wall-time ratio across the full parameter-space scan, and Fig. S11 compares all flash initialization strategies in terms of convergence rate and mean iteration count.

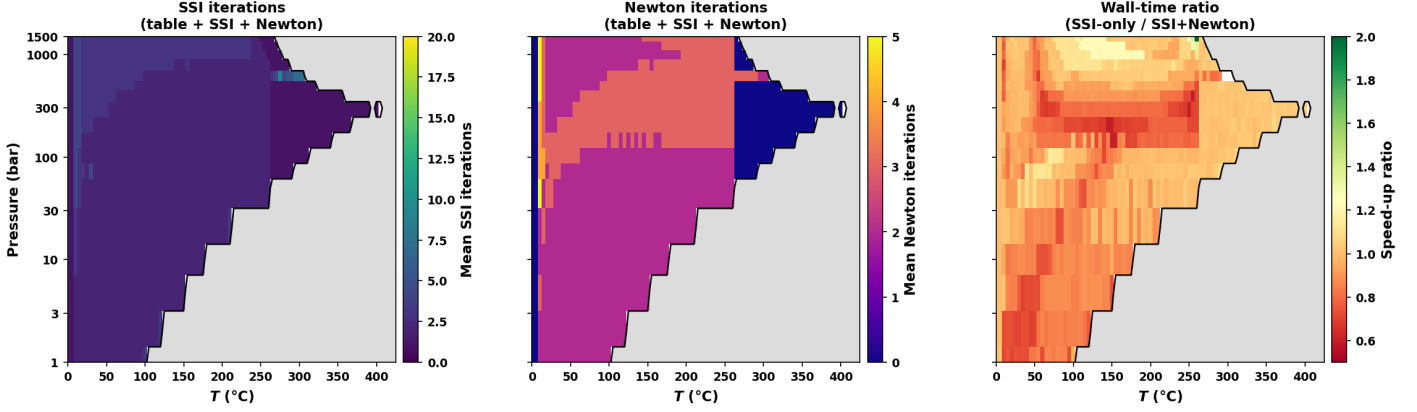


Figure S10: SSI + Newton polish iteration breakdown and wall-time ratio across the parameter-space scan (9,825 two-phase points, averaged over z_{CO_2}). *Left*: mean SSI iteration count before Newton polish is triggered ($\|g\| < 10^{-4}$); the near- CO_2 -critical region ($T \approx 7\text{--}57^\circ\text{C}$) requires the most SSI iterations even with table warm-start. *Center*: mean Newton iteration count; most of the domain converges in 2–3 Newton steps after 1–5 SSI steps, while the critical region (blue) often reverts to SSI before Newton is beneficial. *Right*: wall-time ratio (SSl-only / SSl+Newton); values > 1 (green) indicate that Newton polish is faster; values < 1 (red) indicate that the SSI-only path is faster (primarily near the CO_2 critical point where Newton reversion adds overhead).

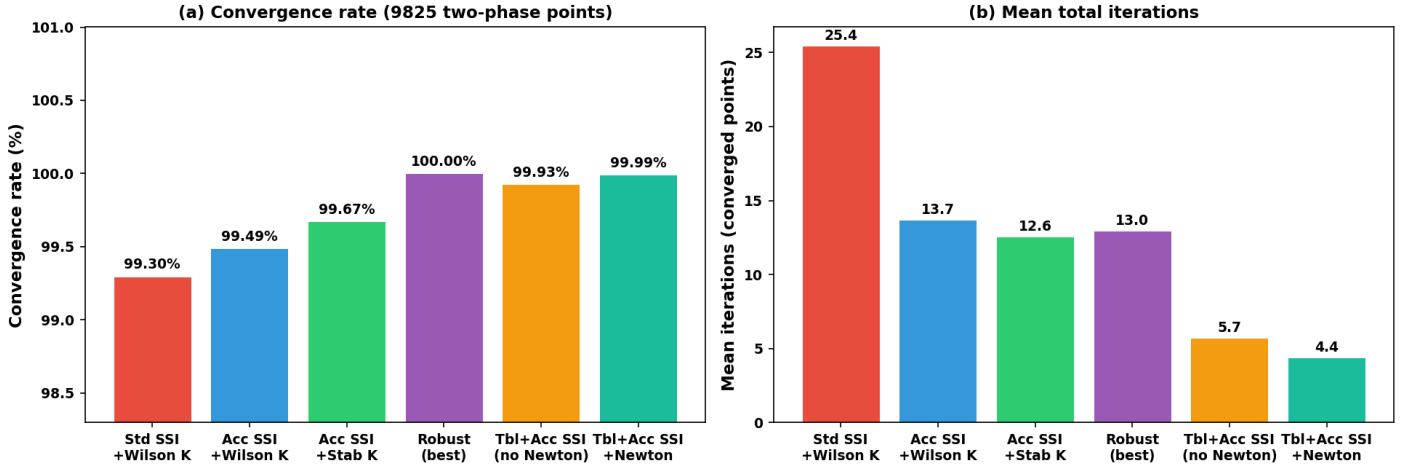


Figure S11: CPA flash strategy comparison across all 9,825 two-phase conditions of the salt-free $\text{CO}_2 + \text{H}_2\text{O}$ parameter-space scan ($T = 0\text{--}425^\circ\text{C}$, $P = 1\text{--}1500$ bar, $z_{\text{CO}_2} = 0.001\text{--}0.999$). (a) Convergence rate; (b) mean total iterations (converged points only). Strategies are ordered from simplest cold-start (standard SSI with Wilson K) to the most optimized (table warm-start with SSI and Newton polish). The final strategy achieves a 73% reduction in mean iterations relative to the standard cold-start baseline.

S6. Reservoir Simulation Details

This section supplements the reservoir simulation section of the main text, providing full details of the numerical scheme (pressure solver, transport discretization, well model), the complete simulation parameter set (Table S6), spatial results and flash performance at $t = 5$ yr (Figs. S12–S13; Table S7).

Numerical scheme

Pressure is solved implicitly at each time step on a uniform Cartesian grid using either a two-point flux approximation (TPFA, Aavatsmark 17) or a lowest-order mixed finite element (MFE, RT₀) scheme [18]. Both were implemented and tested; results shown here use MFE, which avoids the pronounced grid-orientation artefacts of TPFA on Cartesian grids while remaining computationally inexpensive [19, 20, 21]. The solved pressure field yields Darcy velocities that drive explicit first-order upwind advection of the overall CO₂ mole fraction z_{CO_2} . The transport sub-step size is constrained by a CFL condition based on the maximum slope of the molar fractional flow $\partial F/\partial z_{\text{CO}_2}$, estimated by a one-dimensional scan at startup. Production wells are controlled by a fixed bottom-hole pressure (BHP); the injector delivers a fixed volumetric rate.

At each time step, the full Michelsen stability test followed by the two-phase flash is called at every grid cell, using the cell pressure from the previous time step (standard IMPES lagged-pressure). For the eCPA case ($m_s > 0$) the warm-started K-value flash with hierarchical stability fallback is used; for the salt-free case ($m_s = 0$) the CPA warm-started flash is used. NaCl is non-volatile; the equilibrium aqueous molality m_s^{aq} returned by the flash reflects local salting-out and is used directly in the saturation calculation.

We deliberately replace the true compressible volume-rate divergence by the incompressible constraint $\nabla \cdot \mathbf{u} = q$. This neglects buoyancy, compressibility, and the density difference between injected and displaced fluid. We state this limitation explicitly because neglecting it removes the main physical driver of CO₂ plume geometry in real aquifer storage, and readers should not interpret the results as quantitative predictions. The purpose is solely to confirm that the stability and flash routines converge reliably and efficiently at every cell of a 50×50 grid over 300 time steps spanning 5 years.

Simulation setup

Two simulations are run on identical grids with identical flow parameters, differing only in brine salinity:

- **Case A (CPA):** salt-free brine, $m_s = 0 \text{ mol kg}^{-1}$
- **Case B (eCPA):** saline brine, $m_s = 4 \text{ mol kg}^{-1}$ NaCl

The domain is a 100×100 m² square of 10 m thickness discretized onto a 50×50 Cartesian grid ($N = 2500$ cells) with homogeneous permeability of 100 mD and porosity $\phi = 0.20$. A five-spot well pattern is used: a CO₂ injector at the domain center and four BHP-controlled producers at the corners (BHP = 150 bar). CO₂ is injected at 20% of total pore volume per year. The simulation runs for 5 years ($T = 77^\circ\text{C}$) with 300 pressure time steps. Full simulation parameters are listed in Table S6.

Table S6: Reservoir simulation parameters.

Parameter	Symbol	Value
<i>Grid and geometry</i>		
Domain size (x, y)	$L_x \times L_y$	100×100 m
Domain thickness	L_z	10 m
Grid cells (x, y)	$N_x \times N_y$	50×50
<i>Rock properties</i>		
Permeability (isotropic, uniform)	K	100 mD
Porosity	ϕ	0.20
<i>Relative permeability (Corey)</i>		
Connate water saturation	S_{wc}	0.15
Residual gas saturation	S_{gr}	0.05
End-point k_{rg}	k_{rg}^0	0.80
End-point k_{rw}	k_{rw}^0	1.00
Corey exponent (CO ₂ -rich phase)	n_g	2.0
Corey exponent (aqueous phase)	n_w	2.5
<i>Fluid properties</i>		
Temperature (isothermal)	T	77 °C (350 K)
CO ₂ -rich phase viscosity	μ_g	5×10^{-5} Pa s
Aqueous phase viscosity	μ_w	4.5×10^{-4} Pa s
<i>Well and injection parameters</i>		
Well pattern	—	five-spot
Injector location	—	domain center
Producer locations	—	four corners
Wellbore radius	r_w	0.10 m
Producer BHP	P_{BHP}	150 bar
Injection rate	—	20% PVI yr ⁻¹
<i>Simulation control</i>		
Total simulation time	$t_{\max}$	5 yr
Number of pressure time steps	N_t	300
Pressure solver	—	MFE (RT ₀)
Transport scheme	—	first-order upwind, CFL-limited
<i>Cases</i>		
Case A (CPA)	m_s	0 mol kg ⁻¹ (salt-free)
Case B (eCPA)	m_s	4 mol kg ⁻¹ NaCl

Results

Figure S12 shows the final-time ($t = 5$ yr) spatial fields for both cases: the CO₂-rich saturation S_g (top row), the dissolved CO₂ molality m_{CO_2} in the aqueous phase (middle row), and the pressure field (bottom row). The CO₂ saturation maps are strikingly similar between the two cases. This is physically expected: at a high injection rate of 20% PVI yr⁻¹, advection strongly dominates over

dissolution-driven transport. The CO_2 displacement front is governed primarily by the mobility ratio and the imposed pressure gradient, not by whether the equilibrium CO_2 solubility in the aqueous phase is slightly reduced by salt. Given that CO_2 solubility in water is already low—on the order of $1\text{--}2\text{ mol kg}^{-1}$ under typical storage conditions—the further reduction from salting-out has negligible impact on the high-rate displacement dynamics on these time scales.

The dissolved CO_2 maps (middle row) tell a different story. Inside the plume the aqueous phase is everywhere CO_2 -saturated at the local equilibrium value: $m_{\text{CO}_2} \approx 1.25\text{ mol kg}^{-1}$ for the salt-free case and only $m_{\text{CO}_2} \approx 0.63\text{ mol kg}^{-1}$ at $m_s = 4\text{ mol kg}^{-1}$ —a reduction of approximately a factor of two, consistent with the salting-out effect seen in the validation data. The white region around the central injector marks cells where the brine has been fully expelled into the CO_2 -rich phase (complete drying out, $S_g \rightarrow 1$), despite a 15% connate/residual water saturation. Correctly capturing this drying-out regime requires a robust stability and flash model that fully captures the mutual solubilities of CO_2 and H_2O .

No flash failures were recorded in either simulation across all 300 time steps and all 2500 grid cells, confirming the reliability of the hierarchical stability and flash approach.

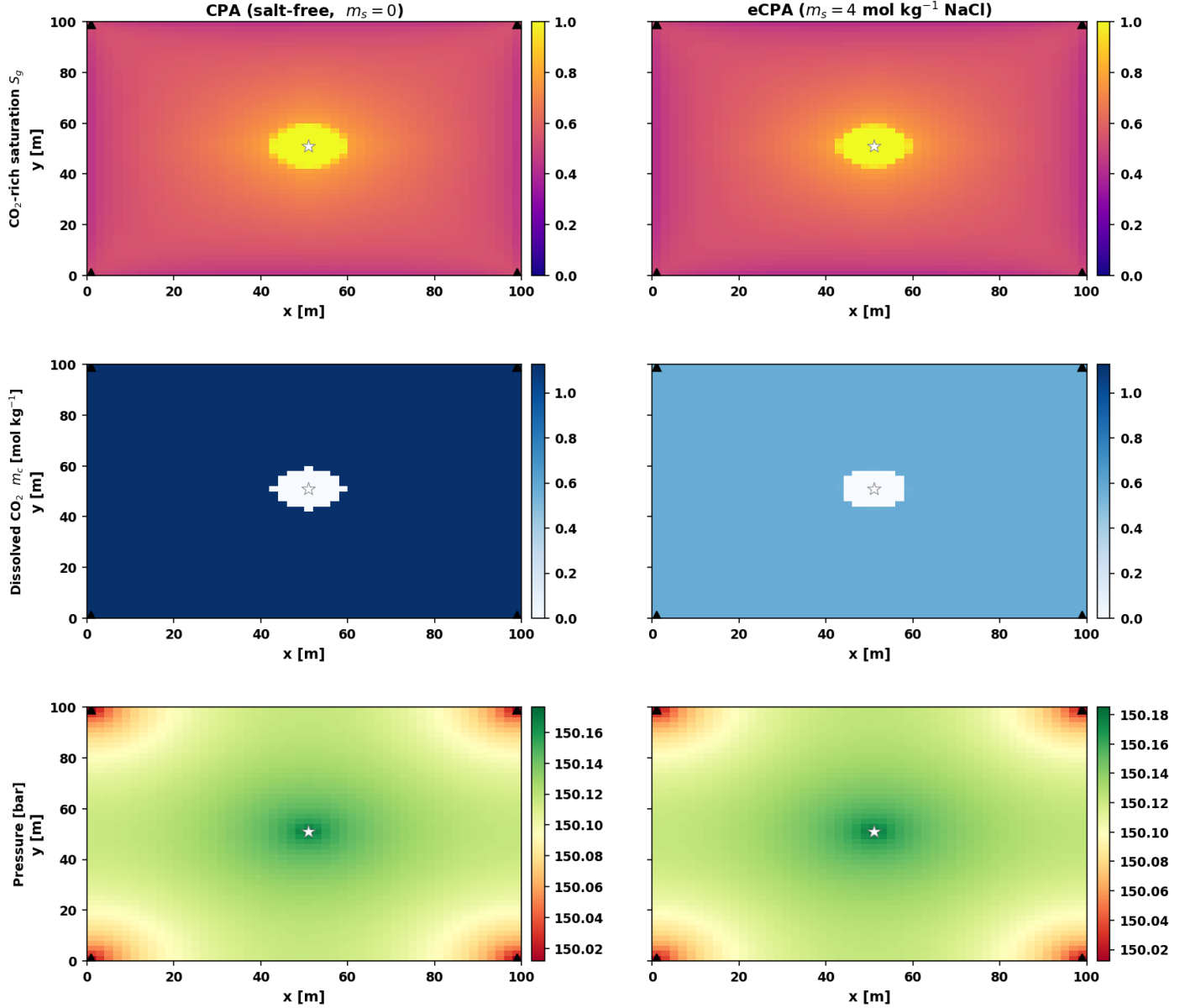


Figure S12: Final-state ($t = 5 \text{ yr}$) spatial fields for the salt-free CPA case ($m_s = 0$, left column) and the saline eCPA case ($m_s = 4 \text{ mol kg}^{-1} \text{ NaCl}$, right column). *Top row:* CO₂-rich phase saturation S_g . The plumes are nearly identical, reflecting advection-dominated displacement at $20\% \text{ PVI yr}^{-1}$. *Middle row:* dissolved CO₂ molality m_{CO_2} in the aqueous phase. Inside the plume the aqueous phase is CO₂-saturated at the eCPA equilibrium value: $\approx 1.25 \text{ mol kg}^{-1}$ (salt-free) vs. $\approx 0.63 \text{ mol kg}^{-1}$ ($m_s = 4 \text{ mol kg}^{-1}$), a factor-of-two reduction from the salting-out effect. The white region near the central injector (white star) marks cells where brine has been completely expelled into the CO₂-rich phase (drying-out). *Bottom row:* pressure field; the $\approx 0.15 \text{ bar}$ gradient across the 100 m domain reflects the small pressure drop required to drive $20\% \text{ PVI yr}^{-1}$ through 100 mD rock with BHP-controlled producers at 150 bar . Black triangles: producers.

While the saturation fields are similar, the implications for long-term CO₂ storage are markedly different. Solubility trapping—the dissolution of CO₂ into the formation brine—is a key mechanism for storage permanence, because dissolved CO₂ cannot escape as a buoyant supercritical phase. Figure S13 shows the fraction of the in-situ CO₂ inventory that is dissolved in brine (solubility trapping, solid lines) versus residing in the CO₂-rich phase (structural trapping, dashed lines) as a function of time

for both cases. The gap between the CPA and eCPA curves is clear and persistent throughout: at $t = 5$ yr the solubility trapping fraction is roughly a factor of two lower for the saline case, directly reflecting the reduction in equilibrium CO_2 solubility from the salting-out effect. This demonstrates that brine salinity has a quantitatively significant impact on long-term storage permanence even when the short-term displacement dynamics appear unaffected.

Flash performance in the simulator

At every time step, the stability test and flash are called at each of the $N = 2500$ grid cells. Cells identified as trivially single-phase bypass the flash entirely; the remaining two-phase candidates are dispatched in parallel across all available CPU cores. As the plume grows, an increasing fraction of the domain becomes two-phase until essentially all cells require a flash call (peak ≈ 2500 calls per step). Performance numbers for both simulations are summarized in Table S7; the flash accounts for the dominant fraction of total step cost throughout both runs, confirming that the algorithmic efficiency of the warm-started flash directly governs simulation throughput. Interestingly, the eCPA flash achieves a higher throughput than CPA (14,270 vs. 8,840 calls s^{-1}), even though the eCPA EoS involves more unknowns and is inherently more expensive per iteration. While one would expect CPA to be the faster of the two in general, the solution-table warm start appears to provide particularly accurate initial guesses for the eCPA system at the temperature, pressure, and salinity conditions of these simulations ($T = 77^\circ\text{C}$, $P \approx 150$ bar, $m_s = 4$ mol kg^{-1}), resulting in fewer SSI iterations to convergence than for the salt-free CPA case at otherwise identical conditions.

Table S7: Aggregate flash performance for the 5-year simulations (50×50 grid, $T = 77^\circ\text{C}$, $P \approx 150$ bar, 300 time steps, 48 CPU cores). Total wall time: 91 s (CPA) and 63 s (eCPA).

Metric	CPA ($m_s = 0$)	eCPA ($m_s = 4$)
Grid cells	2500	2500
Time steps	300	300
Mean two-phase cells / step	2219 (max 2479)	2213 (max 2479)
Mean flash wall time / step (s)	0.247	0.151
Flash fraction of step cost	87%	80%
Flash throughput (calls s^{-1})	8,840	14,270

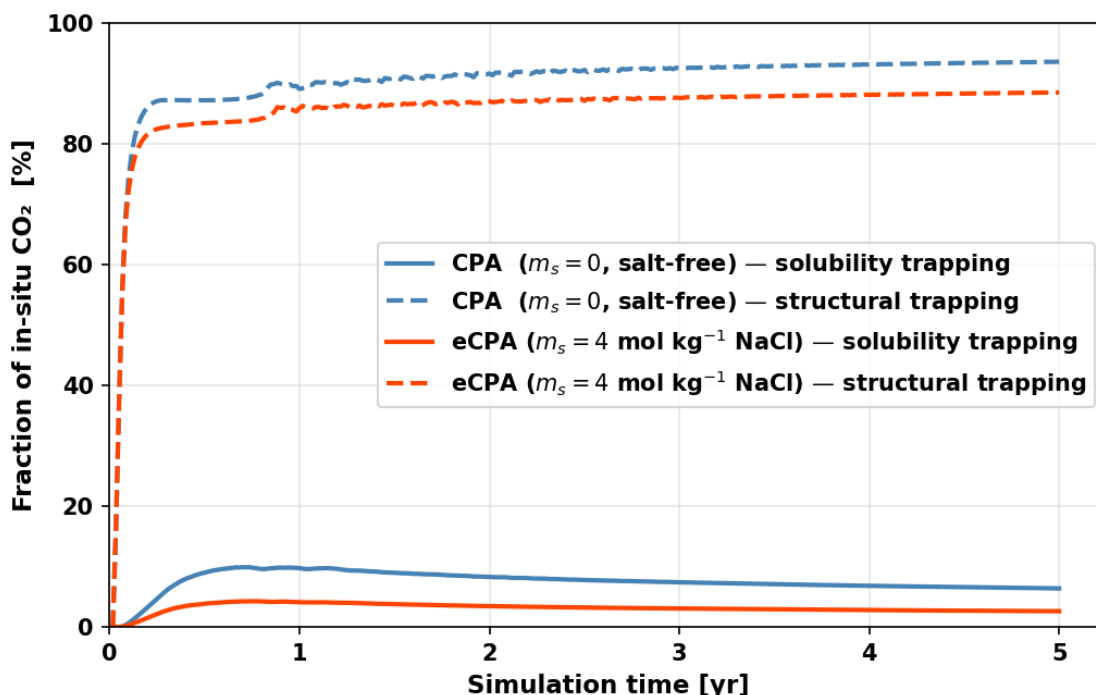


Figure S13: Fraction of total in-situ CO_2 inventory undergoing solubility trapping (dissolved in the aqueous phase, solid lines) versus structural trapping (residing in the CO_2 -rich phase, dashed lines) as a function of time. Blue: salt-free brine (CPA, $m_s = 0$); red: saline brine (eCPA, $m_s = 4 \text{ mol kg}^{-1} \text{ NaCl}$). The persistent factor-of-two gap in solubility trapping between the two cases reflects the reduction in equilibrium CO_2 solubility due to the salting-out effect, which has negligible impact on short-term displacement dynamics but is critical for long-term storage permanence.

References

- [1] F. M. Coelho, L. F. M. Franco, A. Firoozabadi, Phase equilibria of CO_2 -water and CO_2 -brine at high temperatures: From Monte Carlo simulations to the equation of state, *Ind. Eng. Chem. Res.* 64 (2025) 8492–8505. doi:10.1021/acs.iecr.5c00134.
- [2] H. Guo, Y. Huang, Y. Chen, Q. Zhou, Quantitative Raman spectroscopic measurements of CO_2 solubility in NaCl solution from 273.15 to 473.15 K at $p = 10$ to 100 MPa, *J. Chem. Eng. Data* 61 (2015) 466–474. doi:10.1021/acs.jced.5b00799.
- [3] B. Rumpf, H. Nicolaisen, C. Öcal, G. Maurer, Solubility of carbon dioxide in aqueous solutions of sodium chloride: Experimental results and correlation, *J. Solution Chem.* 23 (1994) 431–448. doi:10.1007/BF00973113.
- [4] S. Bando, F. Takemura, M. Nishio, E. Hihara, M. Akai, Solubility of CO_2 in aqueous solutions of nacl at (30 to 60) °c and (10 to 20) mpa, *Journal of Chemical & Engineering Data* 48 (2003) 576–579.
- [5] S. Chabab, P. Théveneau, C. Coquelet, J. Corvisier, P. Fossion, Measurements and predictive models of high-pressure H_2S solubility in brine (NaCl aqueous solution) and mixed gas ($\text{CO}_2 + \text{H}_2\text{S}$) solubilities in various aqueous solutions at high pressures, *Int. J. Greenhouse Gas Control* 93 (2020) 102894. doi:10.1016/j.ijggc.2019.102894, $\text{CO}_2 + \text{NaCl}$ data at $m_s = 6 \text{ mol kg}^{-1}$.

- [6] B. Liu, B. S. Mahmood, E. Mohammadian, A. Khaksar Manshad, N. R. Rosli, M. Ostadhassan, Measurement of solubility of CO₂ in nacl, CaCl₂, MgCl₂ and MgCl₂ + CaCl₂ brines at temperatures from 298 to 373 k and pressures up to 20 mpa using the potentiometric titration method, *Energies* 14 (2021) 7222. URL: <https://www.mdpi.com/1996-1073/14/21/7222>.
- [7] Y. Liu, M. Hou, G. Yang, B. Han, Solubility of CO₂ in aqueous solutions of nacl, kcl, CaCl₂ and their mixed salts at different temperatures and pressures, *The Journal of Supercritical Fluids* 56 (2011) 125–129.
- [8] J. A. Nighswander, N. Kalogerakis, A. K. Mehrotra, Solubilities of carbon dioxide in water and 1 wt. % NaCl solution at pressures up to 10 MPa and temperatures from 80 to 200 °C, *J. Chem. Eng. Data* 34 (1989) 355–360. doi:10.1021/je00057a027.
- [9] S. Takenouchi, G. C. Kennedy, The solubility of carbon dioxide in NaCl solutions at high temperatures and pressures, *Am. J. Sci.* 263 (1965) 445–454. doi:10.2475/ajs.263.5.445.
- [10] W. Yan, S. Huang, E. H. Stenby, Measurement and modeling of CO₂ solubility in NaCl brine and CO₂ + NaCl brine + CH₄ systems, *Int. J. Greenhouse Gas Control* 5 (2011) 1460–1477. doi:10.1016/j.ijggc.2011.08.004.
- [11] S.-X. Hou, G. C. Maitland, J. P. M. Trusler, Measurement and modeling of the phase behavior of the (Carbon Dioxide + Water) and (Carbon Dioxide + Water + NaCl) systems at temperatures from 298.15 to 448.15 K, *J. Supercrit. Fluids* 73 (2013) 87–96. doi:10.1016/j.supflu.2012.11.011.
- [12] D. Koschel, J.-Y. Coxam, L. Rodier, V. Majer, Enthalpy and solubility data of CO₂ in water and NaCl(aq) at conditions of interest for geological sequestration, *Fluid Phase Equilib.* 247 (2006) 107–120. doi:10.1016/j.fluid.2006.06.016.
- [13] H. Messabeb, F. Contamine, P. Cézac, J. P. Serin, E. C. Gaucher, Experimental measurement of CO₂ solubility in aqueous NaCl solution at temperature from 323.15 to 423.15 K and pressure of up to 20 MPa, *J. Chem. Eng. Data* 61 (2016) 3573–3584. doi:10.1021/acs.jced.6b00505.
- [14] V. Savary, G. Berger, M. Dubois, J.-C. Lachapagne, A. Pages, S. Thibaud, M. Lescanne, The solubility of CO₂+H₂S mixtures in water and 2m nacl at 120°C and pressures up to 35mpa, *International Journal of Greenhouse Gas Control* 10 (2012) 123–133.
- [15] H. Zhao, R. Dilmore, D. E. Allen, S. W. Hedges, Y. Soong, S. N. Lvov, Measurement and model predictions of solubility of CO₂ in NaCl aqueous solution at temperatures from 323 to 423 K and pressures up to 40 MPa, *Int. J. Greenhouse Gas Control* 37 (2015) 59–69. doi:10.1016/j.ijggc.2015.02.008.
- [16] W. Wagner, A. Pruß, The IAPWS formulation 1995 for the thermodynamic properties of ordinary water substance for general and scientific use, *Journal of Physical and Chemical Reference Data* 31 (2002) 387–535. doi:10.1063/1.1461829, iAPWS-95; revised release 2016.
- [17] I. Aavatsmark, An introduction to multipoint flux approximations for quadrilateral grids, *Comput. Geosci.* 6 (2002) 405–432.

- [18] G. Chavent, J. Jaffre, Mathematical models and finite elements for reservoir simulation. Studies in mathematics and its applications., Elsevier, North-Holland, 1986.
- [19] J. Moortgat, S. Sun, A. Firoozabadi, Compositional modeling of three-phase flow with gravity using higher-order finite element methods, *Water Resources Research* 47 (2011).
- [20] J. Moortgat, A. Firoozabadi, Higher-order compositional modeling of three-phase flow in 3d fractured porous media based on cross-flow equilibrium, *Journal of Computational Physics* 250 (2013) 425–445.
- [21] J. Moortgat, A. Firoozabadi, Mixed-hybrid and vertex-discontinuous-galerkin finite element modeling of multiphase compositional flow on 3d unstructured grids, *Journal of Computational Physics* 315 (2016) 476–500.